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INSTITUT
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DES SCIENCES

FALCUTÉ DES SCIENCES ET TECHNOLOGIE (FST)

RAPPORT DE TRAVAUX DIRIGÉS I : INTRODUCTION À CISCO PACKET TRACER ET TOPOLOGIES RÉSEAUX

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Niveau : L3

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Introduction

Dans le cadre du développement de compétences en administration du système d'exploitation Linux, la maîtrise de l'interface en ligne de commande et la compréhension de la gestion des environnements graphiques constituent des éléments majeurs. Ce rapport présente les travaux réalisés au cours du TD3, autours des thèmes suivants, Commandes de base, utilitaires, gestion des paquets et bureaux KDE Plasma et XFCE.

Ce TD a pour objectif principale d'acquérir une autonomie pratique dans la gestion quotidienne d'un système Linux. De la manipulation de bas niveau jusqu'à la configuration de l'interface utilisateur.

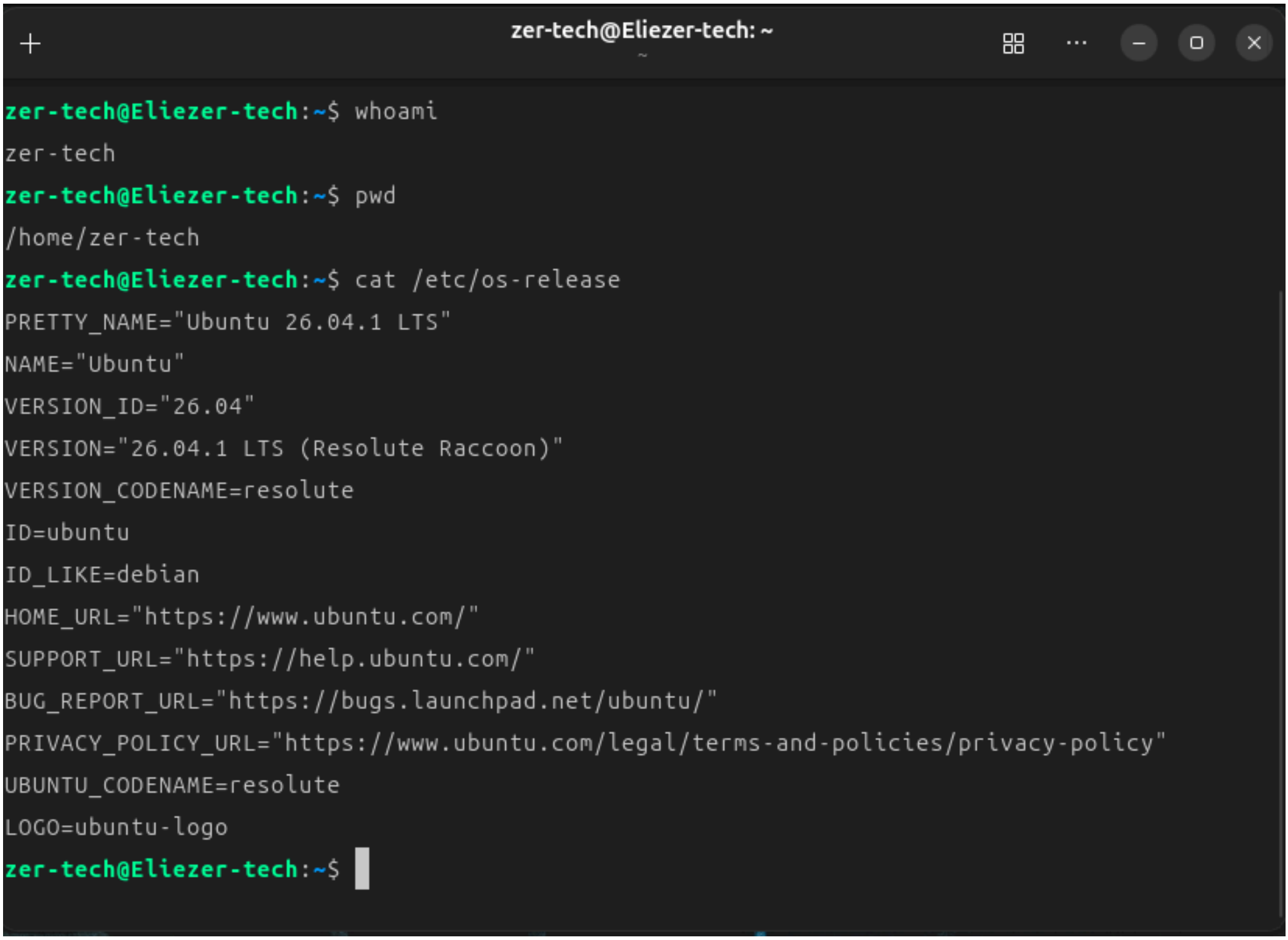
Étant divise en 5 grandes parties, dont la première vise à se repérer et manipuler des fichiers.

EXERCICE 1

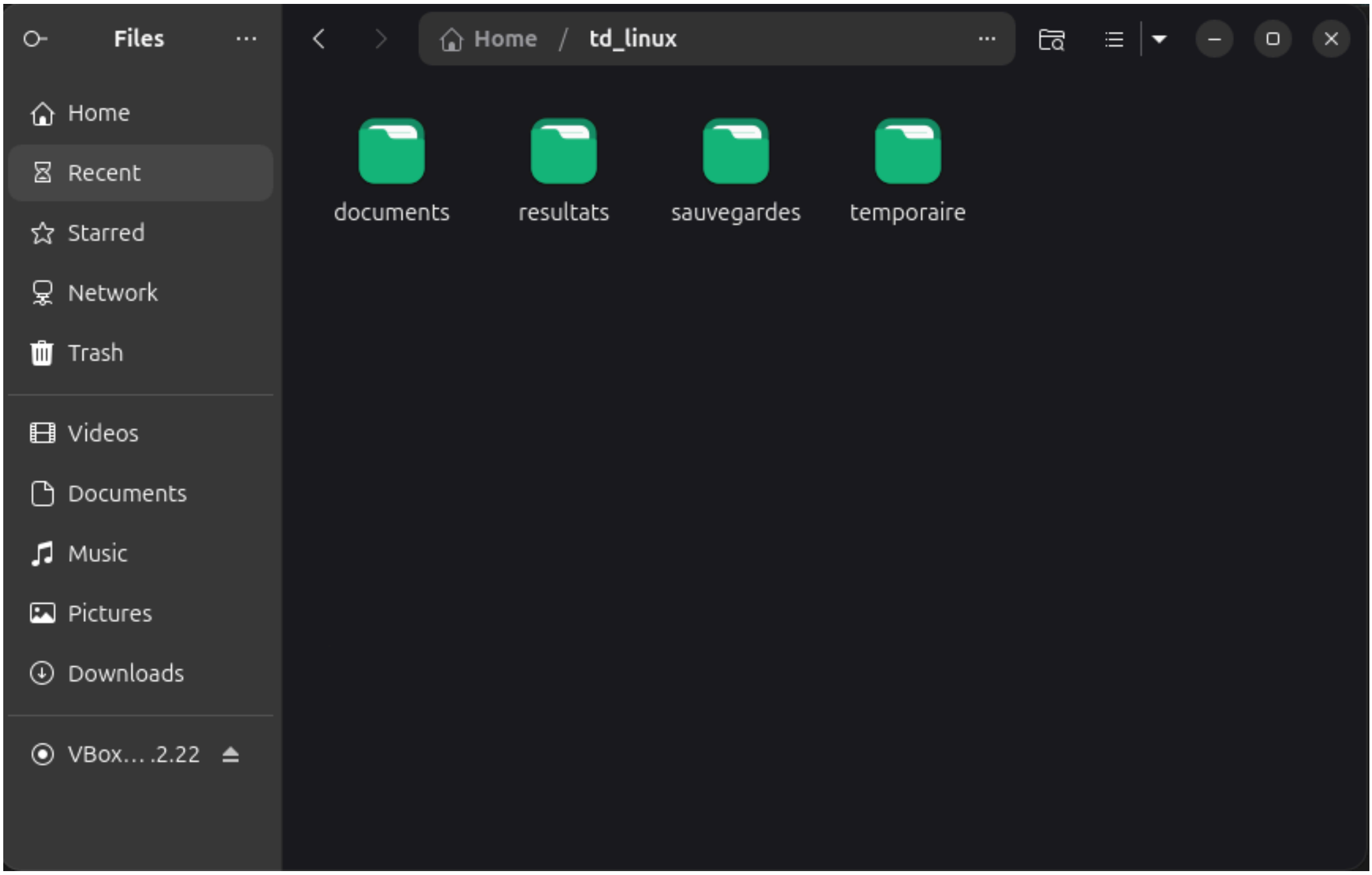
1- Afficher :

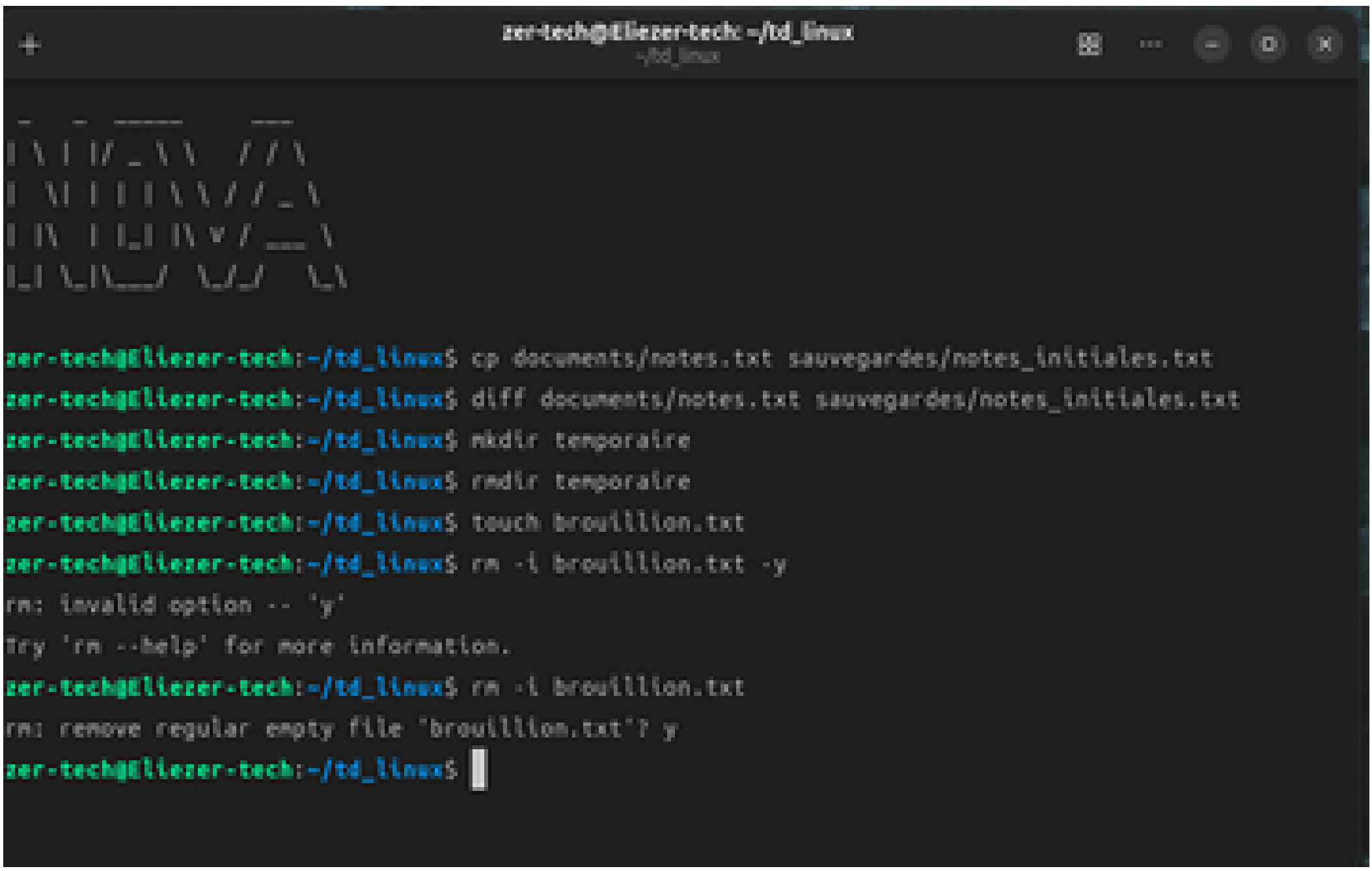
- Le nom de l'utilisateur
- Le chemin du répertoire courant
- Les informations sur la distribution
- La différence entre chemin absolu et chemin relatif.

2- créer l'arborescence *~/td_linux(documents/sauvegardes/résultats)* et l'affiché au format détaillé
3-Créer un fichier *notes.txt,* le copier et le renommer dans *sauvegardes*
4- Créer et supprimer un fichier nommé *temporaire* puis faire la différence entre *rmdir et rm*



Les différents dossiers inclus le dossier temporaire sont bel et bien présent





Sous Linux, la différence réside dans le point de départ utilisé pour localiser un fichier ou un dossier.
Un chemin absolu donne l'adresse complète depuis la racine du système, tandis qu'un chemin relatif donne l'adresse par rapport à l'endroit où vous vous trouvez actuellement (le répertoire de travail).

rmdir (Remove Directory) : Elle sert uniquement à supprimer des répertoires vides. Si le dossier contient le moindre fichier ou sous-dossier, la commande échoue par sécurité pour éviter les pertes de données accidentelles.

rm (Remove) : Elle sert principalement à supprimer des fichiers. Cependant, en lui ajoutant des options (comme -r), elle devient capable de supprimer des dossiers et tout leur contenu (fichiers, sous-dossiers), qu'ils soient vides ou non.

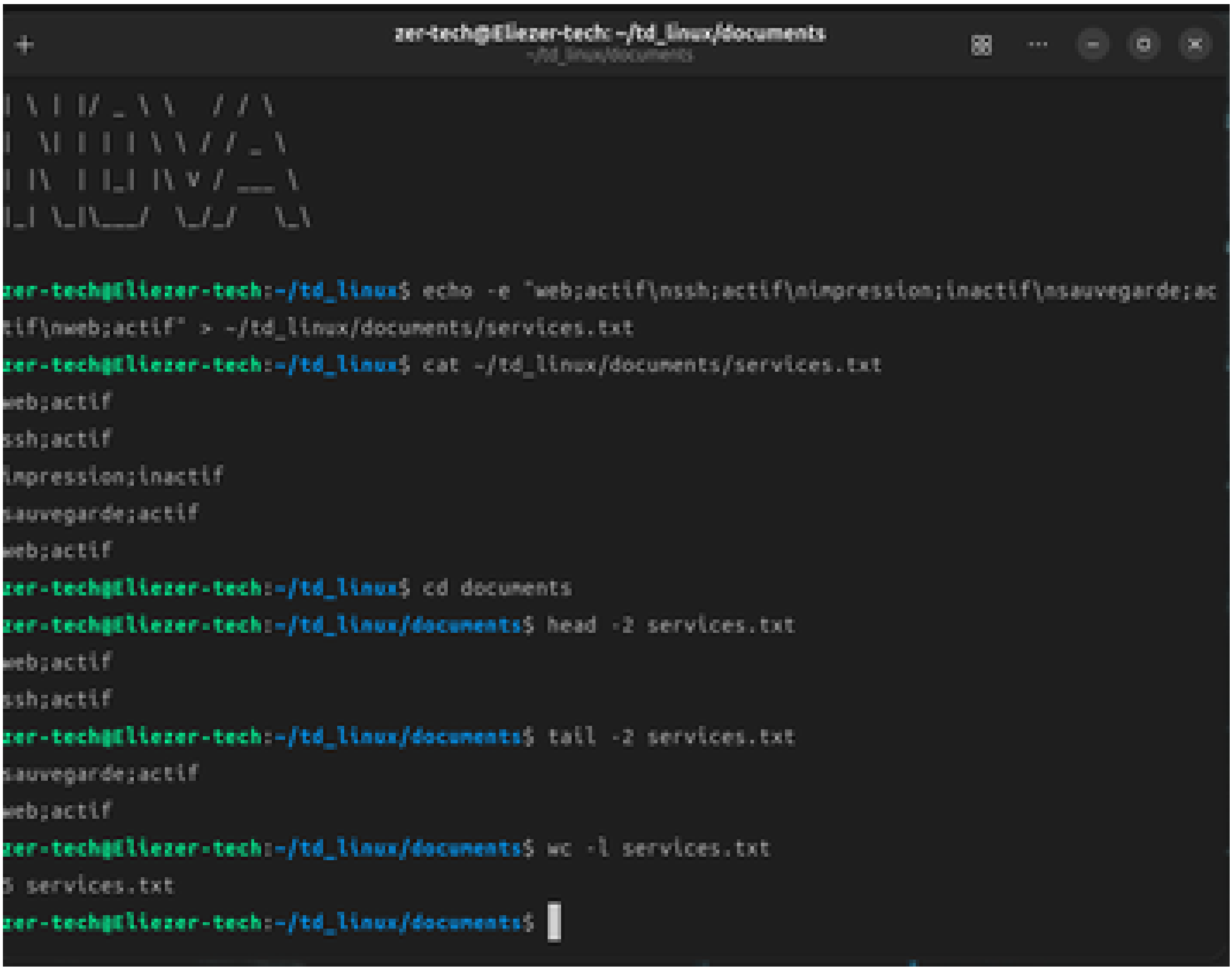
EXERCICE 2

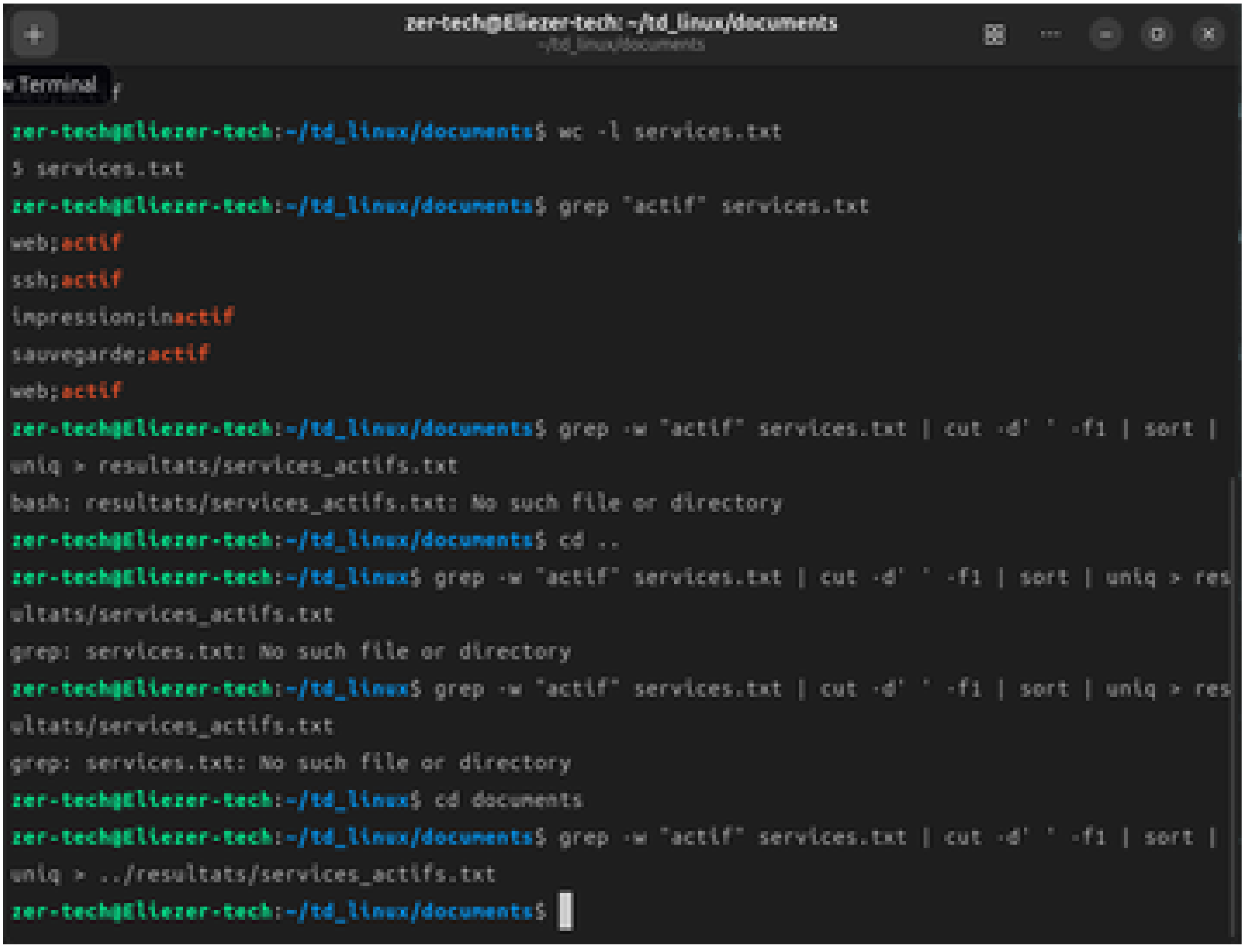
Dans la deuxième partie, vise à exploiter les utilitaires et faire du traitement de texte

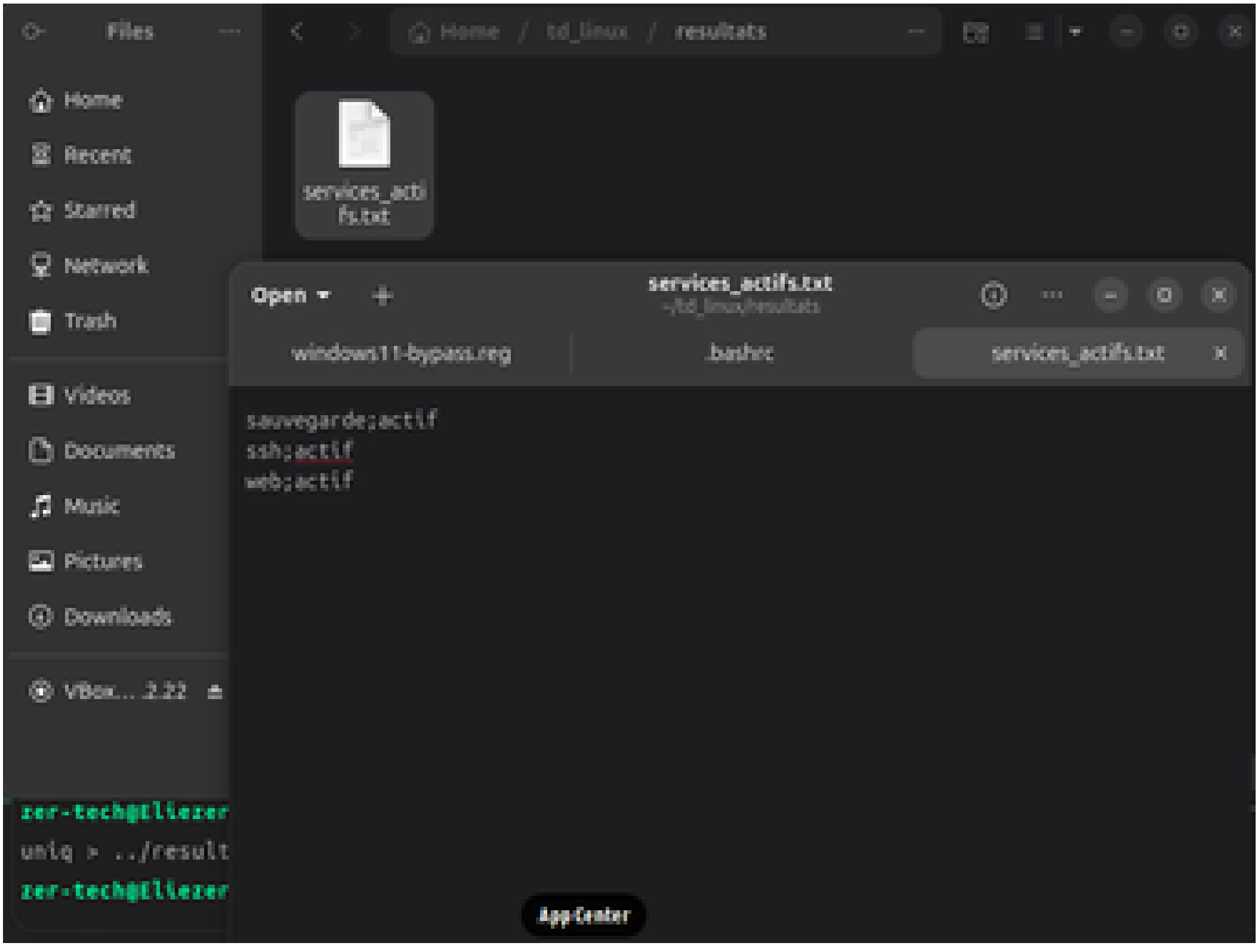
La première tache vise à créer un fichier nommé service.txt et :
1- Afficher

- Le contenu complet du ficher
- Ses deux premières et dernières lignes
- Et son nombre de lignes

2- Extraire uniquement les lignes dont le deuxième champ vaut actif
3- Avec une seule ligne de commande, sélectionner les services actifs, extraire leur noms, les trier et supprimer les doublons, et enregistrer le résultat dans resultats/services_actifs.txt
4- rechercher les fichiers avec l'extension .txt
5- Créer l'archive sauvegardes/documents.tar.gz, afficher son contenu sans l'extraire, puis l'extraire dans un dossier restauration
6- expliquer la différence entre les operateurs > et >>

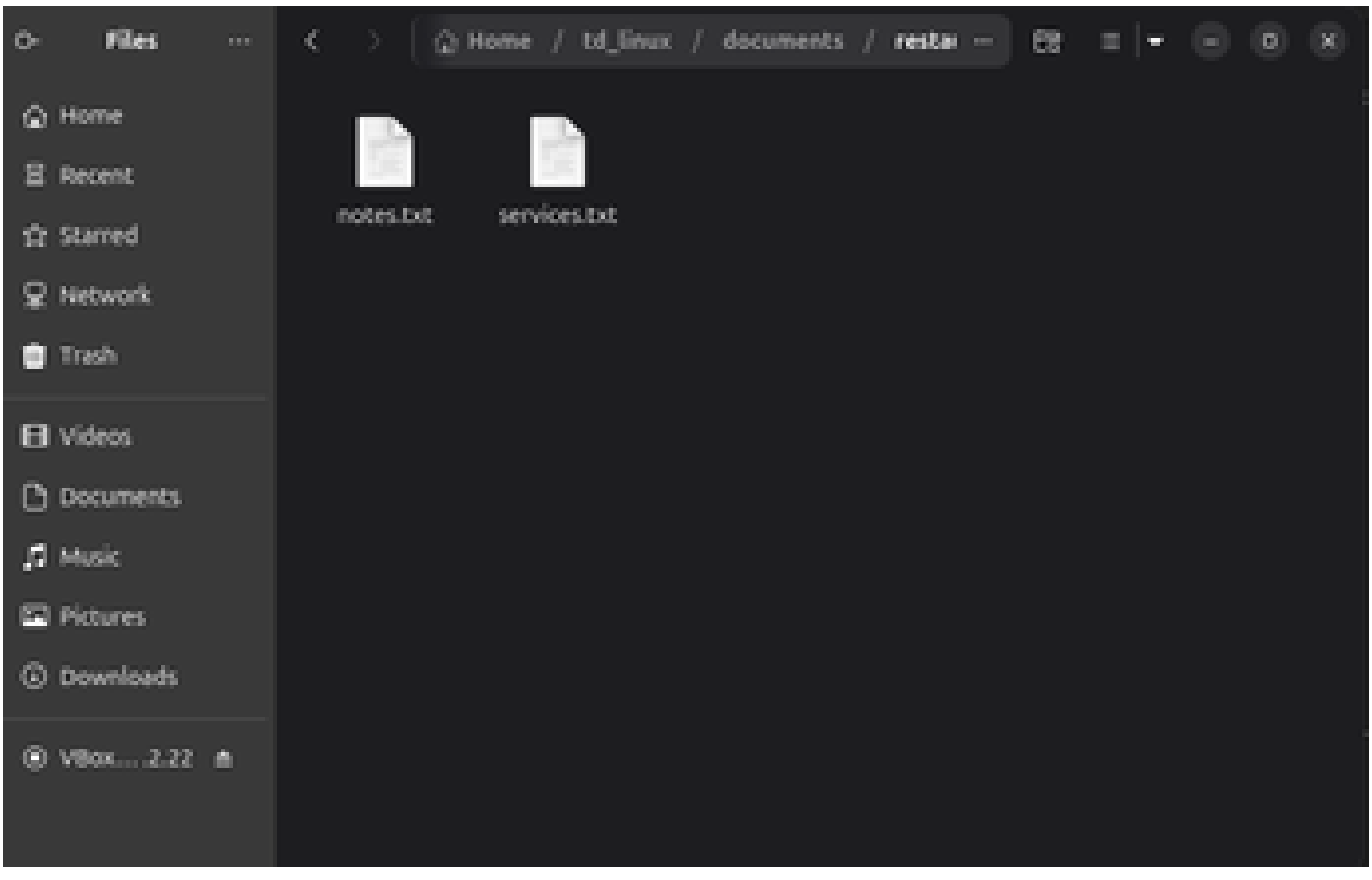



zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
Terminal
zer-tech@Eliezer-tech:~/td_linux/documents$ wc -l services.txt
5 services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ grep "actif" services.txt
web;actif
ssh;actif
impression;inactif
sauvegarde;actif
web;actif
zer-tech@Eliezer-tech:~/td_linux/documents$ grep -w "actif" services.txt | cut -d' ' -f1 | sort |
uniq > resultats/services_actifs.txt
bash: resultats/services_actifs.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux/documents$ cd ..
zer-tech@Eliezer-tech:~/td_linux$ grep -w "actif" services.txt | cut -d' ' -f1 | sort | uniq > res
ultats/services_actifs.txt
grep: services.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux$ grep -w "actif" services.txt | cut -d' ' -f1 | sort | uniq > res
ultats/services_actifs.txt
grep: services.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux$ cd documents
zer-tech@Eliezer-tech:~/td_linux/documents$ grep -w "actif" services.txt | cut -d' ' -f1 | sort |
uniq > ../resultats/services_actifs.txt
zer-tech@Eliezer-tech:~/td_linux/documents$


Files
Home
Recent
Starred
Network
Trash
Videos
Documents
Music
Pictures
Downloads
VBox... 2.22
Home / td_linux / resultats
services_actifs.txt
Open
services_actifs.txt
~/td_linux/resultats
windows11-bypass.reg
.bashrc
services_actifs.txt
sauvegarde;actif
ssh;actif
web;actif
zer-tech@liezer
uniq > ../result
zer-tech@liezer
App Center

```
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -czvf sauvegardes/documents.tar.gz documents/
tar: tar (child): documents: Cannot stat: No such file or directorysauvegardes/documents.tar.gz: C
annot open: No such file or directory

tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -czvf ../sauvegardes/documents.tar.gz .
./
./notes.txt
./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -tzvf ../sauvegardes/documents.tar.gz
drwxrwxr-x zer-tech/zer-tech 0 2026-11-02 16:25 ./
-rw-rw-r-- zer-tech/zer-tech 23 2026-11-02 15:59 ./notes.txt
-rw-rw-r-- zer-tech/zer-tech 60 2026-11-02 16:25 ./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ mkdir restauration
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -xzvf ../sauvegardes/documents.tar.gz -C restaurat
ion/
./
./notes.txt
./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ diff services.txt restauration/services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$
```



La différence entre l'opérateur > et >> réside dans le traitement du contenu existant lors de la redirection du résultat d'une commande vers un fichier texte. L'opérateur > écrase le contenu actuel du fichier, tandis que l'opérateur >> ajoute le contenu à la fin du fichier sans rien effacer.

EXERCICE 3

La troisième partie quant à elle vise a observer le système et les processus

1. Afficher :
   - Expliquer la différence entre *df* et *du* .
   - La date et l'heure.
   - L'espace libre des systèmes de fichiers.
   - L'espace occupé par ~/td_linux .
   - La mémoire disponible.

2. Lancer *sleep* 300 en arrière-plan. Relever immédiatement son PID avec *$!* , puis afficher ce processus avec *ps* .
3. Terminer uniquement ce processus avec *kill* et vérifier son arrêt.
4. Ouvrir *top* , relever un processus et expliquer ses consommations CPU et mémoire. Quitter avec *q* .

```
zer-tech@Eliezer-tech:~$ date
Mon Nov  3 05:14:46 PM UTC 2025
zer-tech@Eliezer-tech:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           676M  2.1M  674M   1% /run
/dev/sda2        29G   12G   16G  42% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
efivarfs        256K   91K  159K  37% /sys/firmware/efi/efivars
tmpfs           1.7G  2.9M  1.7G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/sda1       1.1G  6.4M  1.1G   1% /boot/efi
tmpfs           338M  104K  338M   1% /run/user/1000
/dev/sr0         51M   51M     0 100% /run/media/zer-tech/VBox_GAs_7.2.22
zer-tech@Eliezer-tech:~$ du -sh ~/td_linux
48K     /home/zer-tech/td_linux
zer-tech@Eliezer-tech:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi       1.8Gi       393Mi        64Mi       1.5Gi       1.5Gi
Swap:             0B          0B          0B
```

La commande *df* analyse l'ensemble du système de fichiers , tandis que la commande *du* analyse l'espace occupé par des fichiers ou dossiers précis (vision ciblée)

```
 _   _  _____     ___
| \ | |/ _ \ \   / / \
|  \| | | | \ \ / / _ \
| |\  | |_| |\ V / ___ \
|_| \_|\___/  \_/_/   \_\

zer-tech@Eliezer-tech:~/td_linux/documents$ sleep 300 &
[1] 7367
zer-tech@Eliezer-tech:~/td_linux/documents$ PID=$!
zer-tech@Eliezer-tech:~/td_linux/documents$ ps -p $PID
    PID TTY          TIME CMD
   7367 pts/0    00:00:00 sleep
zer-tech@Eliezer-tech:~/td_linux/documents$ kill $PID
[1]+  Terminated              sleep 300
zer-tech@Eliezer-tech:~/td_linux/documents$ ps -p $PID
    PID TTY          TIME CMD
zer-tech@Eliezer-tech:~/td_linux/documents$ top

top - 17:53:40 up  2:22,  1 user,  load average: 0.17, 0.32, 0.23
Tasks: 224 total,   1 running, 223 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.3 us,  5.9 sy,  0.0 ni, 80.3 id,  0.0 wa,  0.0 hi, 10.5 s
```

zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents

```
Tasks: 224 total,   1 running, 223 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.3 us,  5.9 sy,  0.0 ni, 80.3 id,  0.0 wa,  0.0 hi, 10.5 s
MiB Mem :   3379.9 total,    198.7 free,   1988.5 used,   1504.0 buff
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   1391.4 avai

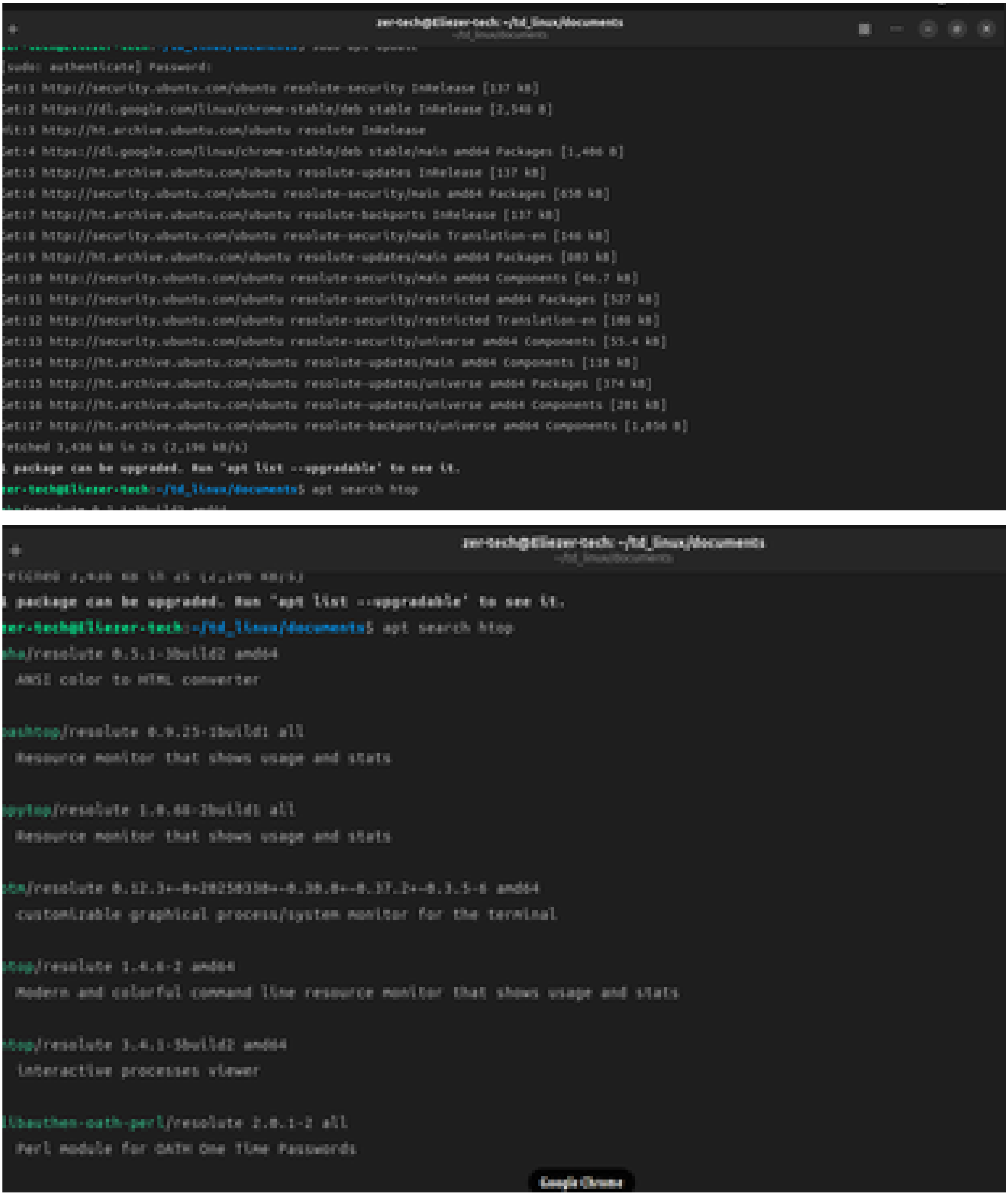
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
   2965 zer-tech  20   0 4117504 586292 183932 S  10.9  14.0
   7305 zer-tech  20   0 1089100 387460 142376 S   9.3  11.2
   4340 root      20   0       0      0      0 I   1.7   0.0
   6312 zer-tech  20   0   52.8g 177452 144148 S   1.3   5.1
   6313 zer-tech  20   0   52.4g 129960 108984 S   0.7   3.8
   6883 root      20   0       0      0      0 I   0.7   0.0
      1 root      20   0   25460  10844  11700 S   0.0   0.5
      2 root      20   0       0      0      0 S   0.0   0.0
      3 root      20   0       0      0      0 S   0.0   0.0
      4 root       0 -20       0      0      0 I   0.0   0.0
      5 root       0 -20       0      0      0 I   0.0   0.0
      6 root       0 -20       0      0      0 I   0.0   0.0
      7 root       0 -20       0      0      0 I   0.0   0.0
      8 root       0 -20       0      0      0 I   0.0   0.0
     10 root       0 -20       0      0      0 I   0.0   0.0
     12 root      20   0       0      0      0 I   0.0   0.0
rer-tech@Eliezer-tech:~/td_linux/documents$
```

files

EXERCICE 4

La partie 4 vise à faire l'installation et la gestion des packets

Sous Linux la différence réside dans l'action effectuée. En effet, tandis que l'une met à jour la liste des logiciels disponibles, l'autre installe concrètement les nouvelles versions. Donc *sudo apt update actualise l'index* des paquets, tandis que *sudo apt upgrade* télécharge et installe les mises à jour logicielles sur la machine.



```
ftools/resolute 3.2.12-1 amd64
  build and search protein and DNA generalized profiles

ser-tech@tlinzer-tech:~/td_linux/documents$ apt show htop
Package: htop
Version: 3.4.1-5build2
Priority: optional
Section: utils
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Daniel Lange <DLange@debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 451 kB
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Homepage: https://htop.dev/
Task: cloud-image, cloud-image, server, ubuntu-server-raspi, kubuntu-desktop, kubuntu-full, lubuntu-desktop, ubuntu-mate-core, ubuntu-mate-
Download-Size: 177 kB
APT-Sources: http://bt.archive.ubuntu.com/ubuntu resolute/main amd64 Packages
Description: interactive processes viewer
 htop is an ncurses-based process viewer similar to top, but it
 allows one to scroll the list vertically and horizontally to see
```

```
ser-tech@tlinzer-tech:~/td_linux/documents$ apt -s install htop
Error: Invalid operation htop
ser-tech@tlinzer-tech:~/td_linux/documents$ apt -s install htop
NOTE: This is only a simulation!
      apt needs root privileges for real execution.
      Keep also in mind that locking is deactivated,
      so don't depend on the relevance to the real current situation!

The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'apt autoremove' to remove it.

Installing:
  htop

Suggested packages:
  lm-sensors

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Inst htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
Conf htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
ser-tech@tlinzer-tech:~/td_linux/documents$
```

```
  lm-sensors

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Need htop (3.4.1-5build2 Ubuntu:24.04/resolute [amd64])
Setf htop (3.4.1-5build2 Ubuntu:24.04/resolute [amd64])
ser-tech@Elsaser-tech:~/Id_Linux/Documents$ sudo apt install htop
The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'sudo apt autoremove' to remove it.

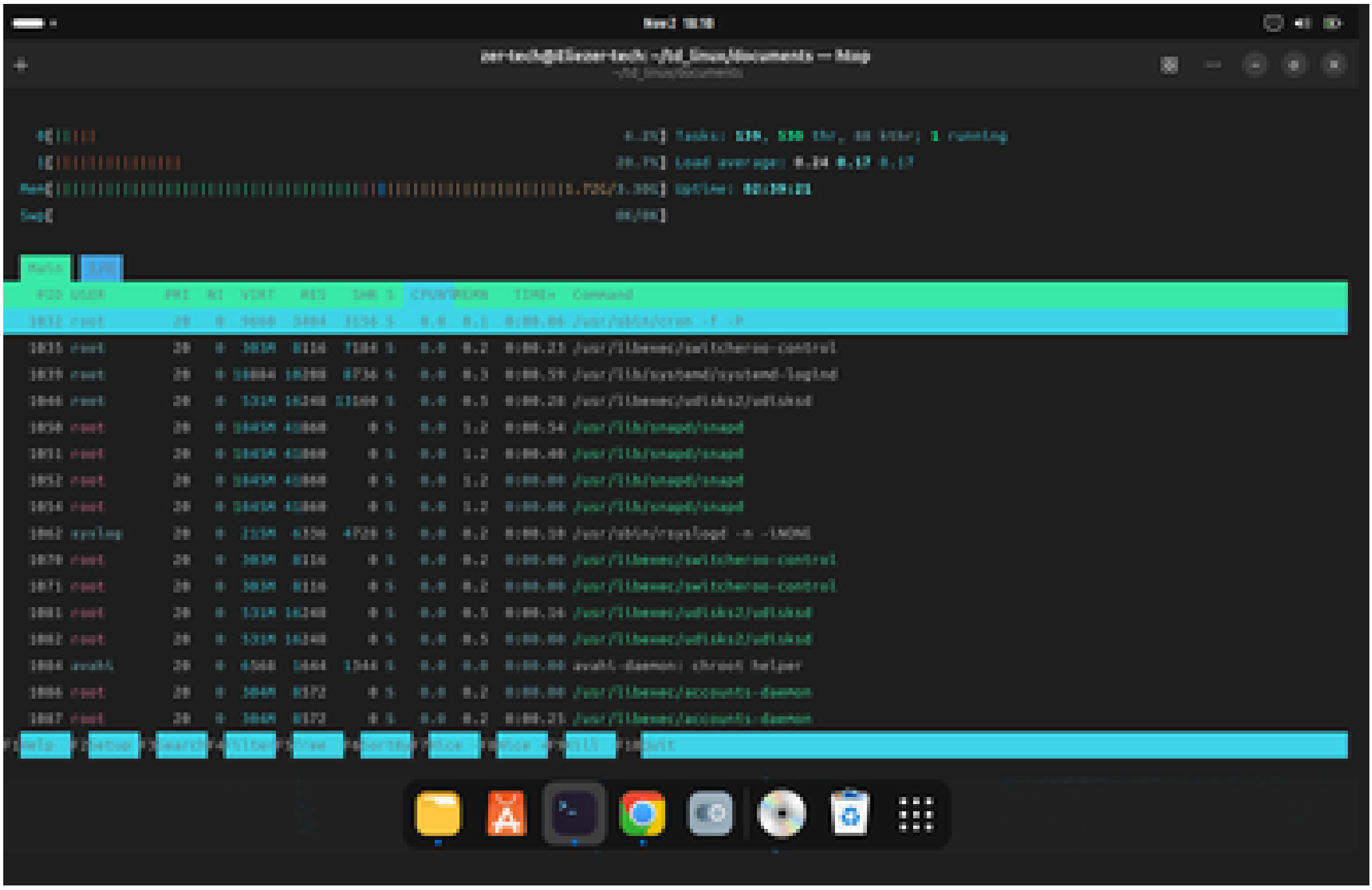
Installing:
  htop

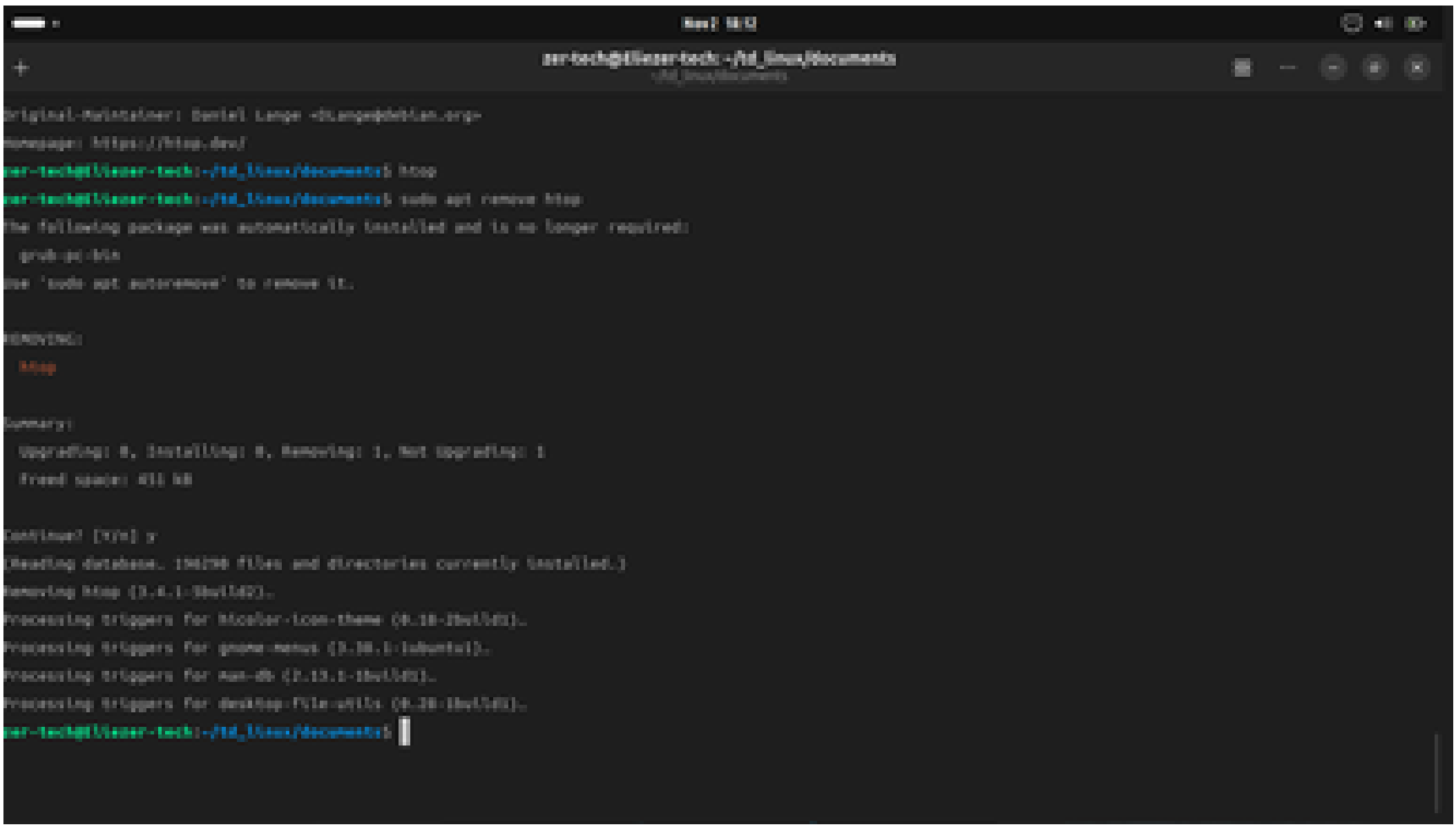
Suggested packages:
  lm-sensors

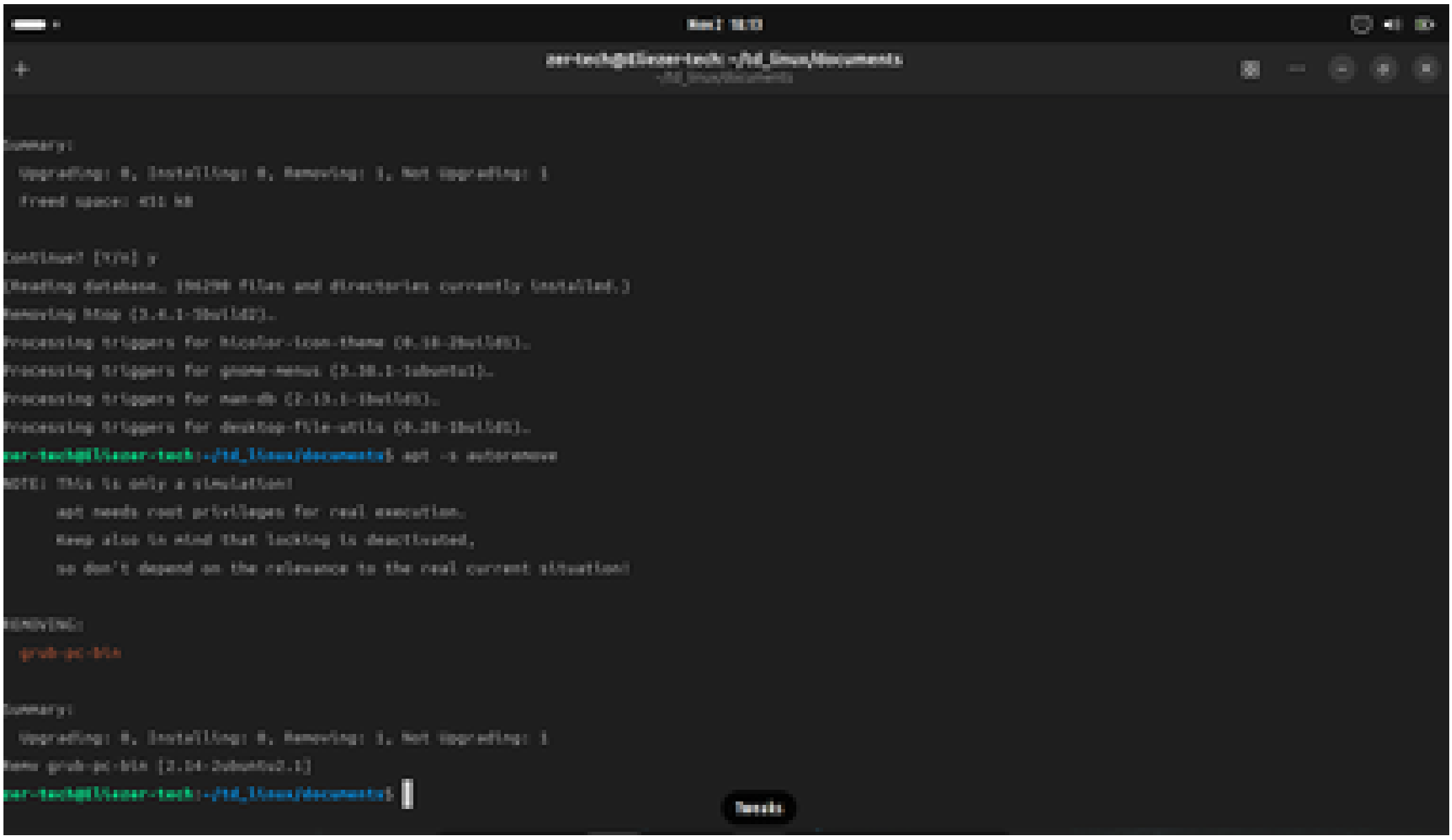
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 177 kB
  Space needed: 451 kB / 24.0 GB available

Get:1 http://th.archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Fetched 177 kB in 1s (249 kB/s)
Selecting previously unselected package htop.
(Reading database ... 196282 files and directories currently installed.)
Preparing to unpack .../htop_3.4.1-5build2_amd64.deb ...
Unpacking htop (3.4.1-5build2) ...
Setting up htop (3.4.1-5build2) ...
Processing triggers for desktop-file-utils (0.28-1build1) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for gnome-menus (3.36.1-1ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
ser-tech@Elsaser-tech:~/Id_Linux/Documents$
```

```
Processing triggers for desktop-file-utils (0.28-1build1) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for gnome-menus (3.36.1-1ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
ser-tech@Elsaser-tech:~/Id_Linux/Documents$ dpkg -s htop
Package: htop
Status: install ok installed
Priority: optional
Section: utils
Installed-Size: 448
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: amd64
Version: 3.4.1-5build2
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Description: interactive processes viewer
 htop is an ncurses-based process viewer similar to top, but it
 allows one to scroll the list vertically and horizontally to see
 all processes and their full command lines.
 .
 Tasks related to processes (killing, renicing) can be done without
 entering their PIDs.
Original-Maintainer: Daniel Lange <DLange@debian.org>
Homepage: https://htop.dev/
ser-tech@Elsaser-tech:~/Id_Linux/Documents$
```


Tasks: 336, 550 thr, 10 kthr; 1 running
Load average: 0.24 0.17 0.17
Uptime: 02:19:21
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1832 root 20 0 9468 3484 3156 S 0.0 0.1 0:00.06 /usr/sbin/cron -f -P
1831 root 20 0 385M 8116 7184 S 0.0 0.2 0:00.23 /usr/libexec/switcheroo-control
1839 root 20 0 18884 10208 8736 S 0.0 0.3 0:00.59 /usr/lib/systemd/systemd-logind
1846 root 20 0 131M 16248 11560 S 0.0 0.5 0:00.28 /usr/libexec/udisks2/udisksd
1850 root 20 0 1845M 45868 0 S 0.0 1.2 0:00.54 /usr/lib/snapd/snapd
1851 root 20 0 1845M 45868 0 S 0.0 1.2 0:00.40 /usr/lib/snapd/snapd
1852 root 20 0 1845M 45868 0 S 0.0 1.2 0:00.09 /usr/lib/snapd/snapd
1854 root 20 0 1845M 45868 0 S 0.0 1.2 0:00.00 /usr/lib/snapd/snapd
1862 syslog 20 0 222M 6156 4720 S 0.0 0.2 0:00.10 /usr/sbin/rsyslogd -n -iNONE
1870 root 20 0 385M 8116 0 S 0.0 0.2 0:00.00 /usr/libexec/switcheroo-control
1871 root 20 0 385M 8116 0 S 0.0 0.2 0:00.00 /usr/libexec/switcheroo-control
1881 root 20 0 131M 16248 0 S 0.0 0.5 0:00.16 /usr/libexec/udisks2/udisksd
1882 root 20 0 131M 16248 0 S 0.0 0.5 0:00.00 /usr/libexec/udisks2/udisksd
1884 avahi 20 0 4568 1604 1544 S 0.0 0.0 0:00.00 avahi-daemon: chroot helper
1886 root 20 0 384M 8372 0 S 0.0 0.2 0:00.00 /usr/libexec/accounts-daemon
1887 root 20 0 384M 8372 0 S 0.0 0.2 0:00.21 /usr/libexec/accounts-daemon
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice -F8Nice +F9Kill F10Quit


zer-tech@tlinzer-tech: ~/td_linux/documents
Original-Maintainer: Daniel Lange <DLange@debian.org>
Homepage: https://htop.dev/
zer-tech@tlinzer-tech:~/td_linux/documents$ htop
zer-tech@tlinzer-tech:~/td_linux/documents$ sudo apt remove htop
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'sudo apt autoremove' to remove it.
REMOVING:
htop
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
Freed space: 411 kB
Continue? [Y/n] y
(Reading database ... 196290 files and directories currently installed.)
Removing htop (3.4.1-5build2) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for gnome-menus (3.36.1-1ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for desktop-file-utils (0.28-1build1) ...
zer-tech@tlinzer-tech:~/td_linux/documents$


Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
Freed space: 471 kB
(Reading database ... 196290 files and directories currently installed.)
Removing htop (3.4.1-5build2) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for desktop-file-utils (0.28-1build1) ...
apt -s autoremove
NOTE: This is only a simulation!
apt needs root privileges for real execution.
Keep also in mind that locking is deactivated,
so don't depend on the relevance to the real current situation!
REMOVING:
grub-pc-bin
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
Remv grub-pc-bin [2.14-2ubuntu2.1]

Exécutons ces commandes

ping google.com # Tester la connectivité réseau

ip a # Affiche les interfaces réseau et IP

history # Affiche l’historique des commandes

cowsay "Bienvenue dans le cours Linux" # Le terminal te parle avec une vache

cmatrix # Effet “Matrix” dans le terminal

neofetch # Affiche les infos système avec logo Linux

journalctl -xe # Consulter les logs système récents

sudo dmesg

sudo dmesg | grep -i memory

date

cal

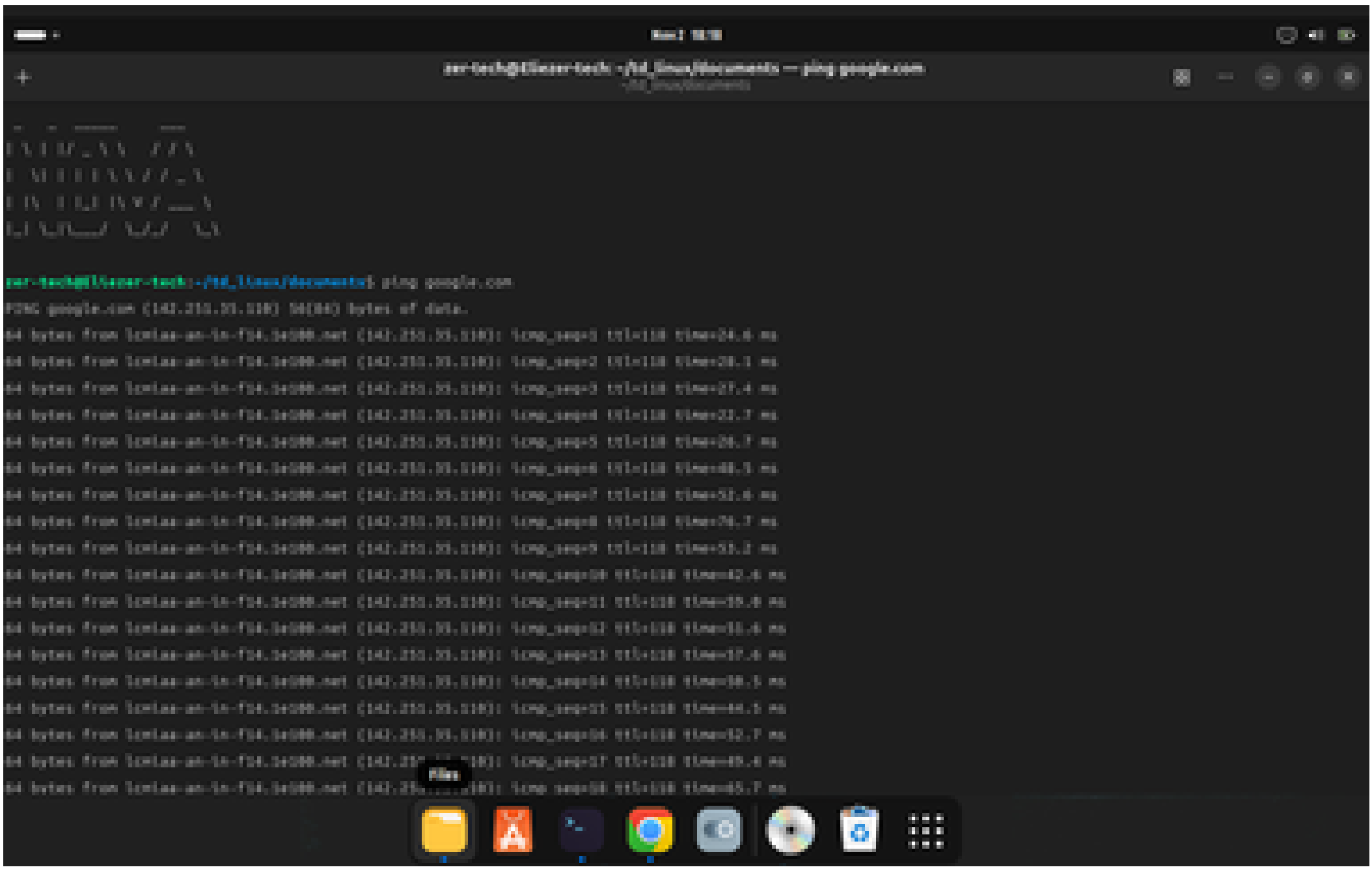
figlet "UBUNTU"# Écrire un grand titre

sl # Écrire un grand titre

seq 1 10 # Écrire un grand titre

seq 2 2 20 # Affiche les nombres pairs de 2 à 20.

shuf -i 1-100 -n 1 # Affiche un nombre aléatoire entre 1 et 100.

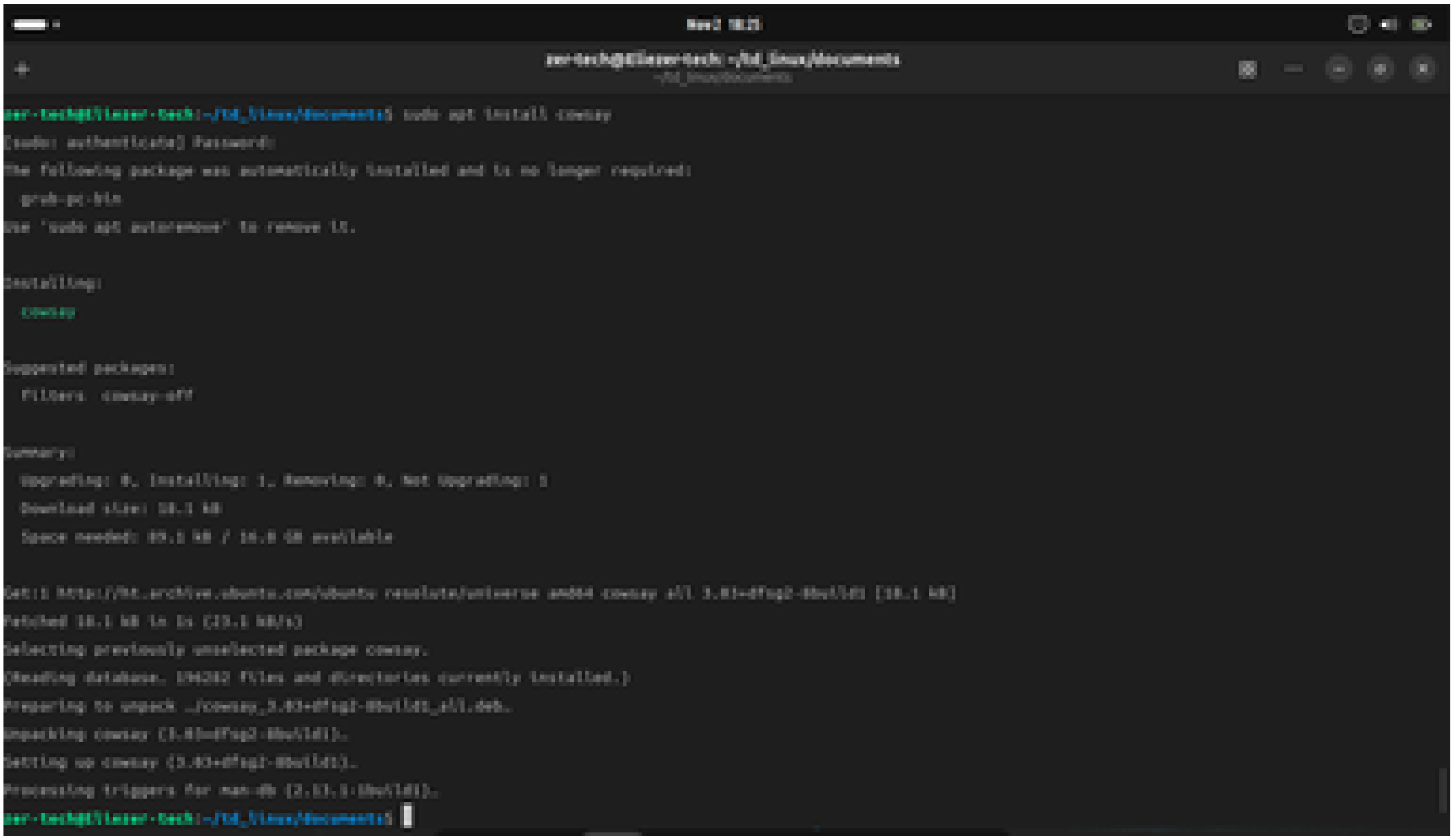


```
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=289 ttl=118 time=51.6 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=290 ttl=118 time=52.6 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=291 ttl=118 time=66.6 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=292 ttl=118 time=71.8 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=293 ttl=118 time=54.7 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=294 ttl=118 time=70.7 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=295 ttl=118 time=71.1 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=296 ttl=118 time=60.3 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=297 ttl=118 time=64.8 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=298 ttl=118 time=58.7 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=299 ttl=118 time=57.7 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=300 ttl=118 time=76.2 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=301 ttl=118 time=60.4 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=302 ttl=118 time=71.2 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=303 ttl=118 time=63.8 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=304 ttl=118 time=60.8 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=305 ttl=118 time=100 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=306 ttl=118 time=100 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=307 ttl=118 time=114 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=308 ttl=118 time=100 ms
^C
--- google.com ping statistics ---
309 packets transmitted, 303 received, 1.623308 packet loss, time 307932ms
rtt min/avg/max/mdev = 22.215/48.862/132.415/15.681 ms
zer-tech@Eleazer-Tech:~/Old_Linux/Documents$
```

```
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=304 ttl=118 time=60.8 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=305 ttl=118 time=100 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=306 ttl=118 time=100 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=307 ttl=118 time=114 ms
64 bytes from lon16aa-am-in-f14.1e100.net (142.251.35.110): icmp_seq=308 ttl=118 time=100 ms
^C
--- google.com ping statistics ---
309 packets transmitted, 303 received, 1.623308 packet loss, time 307932ms
rtt min/avg/max/mdev = 22.215/48.862/132.415/15.681 ms
zer-tech@Eleazer-Tech:~/Old_Linux/Documents$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:57:9a:52 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 76296sec preferred_lft 76296sec
    inet6 fd57:e25a:f937:2:a00:27ff:fe57:9a52/64 scope global dynamic noprefixroute
       valid_lft 86371sec preferred_lft 14371sec
    inet6 fe80::a00:27ff:fe57:9a52/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
zer-tech@Eleazer-Tech:~/Old_Linux/Documents$
```

```
    \valid_lft forever preferred_lft forever
ser-tech@Elixier-tech:~/Id_Linux/Documents$ history
    1  sudo apt update
    2  sudo apt upgrade -y
    3  sudo apt install net-tools curl wget git build-essential -y
    4  sudo apt upgrade -y
    5  sudo apt install net-tools curl wget git build-essential -y
    6  sudo dpkg --configure -a
    7  sudo apt install net-tools curl wget git build-essential -y
    8  man
    9  cd /home
   10  cd ..
   11  cd ~
   12  whoami
   13  pwd
   14  sudo apt install gnome-tweaks -y
   15  sudo apt install gnome-shell-extensions
   16  gsettings set org.gnome.shell.extensions.dash-to-dock click-actions 'minimize'
   17  gsettings set org.gnome.shell.extensions.dash-to-dock click-actions 'minimize'
   18  gsettings set org.gnome.shell.extensions.dash-to-dock click-actions 'minimize'
   19  gsettings set org.gnome.shell.extensions.dash-to-dock click-action 'minimize'
   20  sudo apt install flatpak
   21  flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpakrepo
   22  flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpakrepo
   23  sudo apt install gnome-shell-extensions gnome-shell-extension-manager
```

```
   95  kill $PID
   96  ps -p $PID
   97  sleep 300 &
   98  PID=$!
   99  ps -p $PID
  100  kill $PID
  101  ps -p $PID
  102  top
  103  sudo apt update
  104  apt search htop
  105  apt show htop
  106  apt policy htop
  107  apt -s install htop
  108  apt install htop
  109  apt -S install htop
  110  apt -s install htop
  111  sudo apt install htop
  112  dpkg -s htop
  113  htop
  114  sudo apt remove htop
  115  apt -s autoremove
  116  ping google.com
  117  ip a
  118  history
```



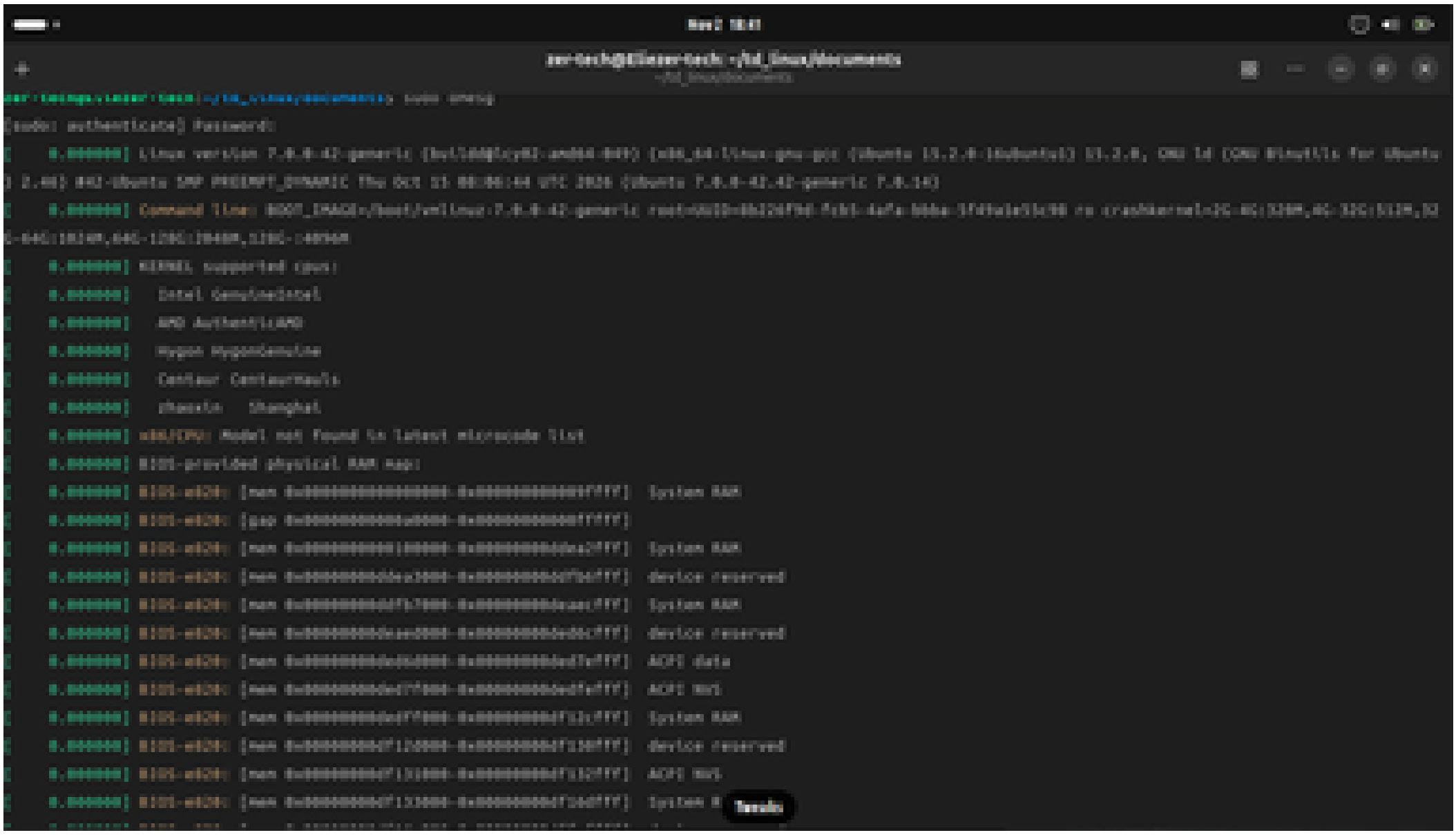


nous avons utilisé Fastfetch au lieu de Neofetch car le packet était absent

```
                    .',:clooo:  .:looooo:.                 ....................
                 .;looooooooc  .oooooooooo'                OS: Ubuntu 26.04.1 LTS (Resolute Raccoon) x86_64
              .;looooooool:,''.  :ooooooooooc               Host: VirtualBox (1.2)
            ;looool;.         'ooooooooooo,.               Kernel: Linux 7.0.0-42-generic
          ;clool'              .cooooooo.  ,,              Uptime: 2 mins
            ...                  ......  .:oo,.            Packages: 1589 (dpkg), 16 (snap)
       .;clol:,.                        .looool'           Shell: bash 5.3.9
     :ooooooooo.                         'ooool           Display (VirtualBox): 1280x800, 60 Hz
   'ooooooooooo.                          loooo.          DE: GNOME 50.1
   'ooooooooool                           coooo.          WM: Mutter (Wayland)
   ,looooooooo.                          .looooo.         WM Theme: Yaru
     .,clc'.                             :ooooc           Theme: Yaru [GTK2/3/4]
        ...                             ,ooool.           Icons: Reversal-green-dark [GTK2/3/4]
     .cooooc.              ..',,'.  .cooo.                Font: Ubuntu Sans 11 @wdth=75 wght=400 (Ultracold, Condensed) [GTK2/3/4]
       ;ooooo:.          ;oooooooc.  :l.                  Cursor: Yaru (24px)
        .coooooc,..      coooooooooo.                     Terminal: Ptyxis 50.1
          .:ooooooolc:. .ooooooooooo'                     Terminal Font: Ubuntu Sans Mono (11pt)
            .'cooooooc.  ,oooooooooo                      CPU: Intel(R) Core(TM) i5-10500U (2) @ 2.21 GHz
                ..'::c'  .:looooc:'                       GPU: VMware SVGA II Adapter
                                                          Memory: 1.22 GiB / 3.20 GiB (37%)
                                                          Swap: Disabled
                                                          Disk (/): 11.33 GiB / 28.33 GiB (40%) - ext4
                                                          Disk (/run/media/ser-tech/VBox_GAs_7.2.22): 58.74 MiB / 58.74 MiB (100%) - iso9660 [Read-only]
                                                          Local IP (enp0s3): 10.0.2.15/24
                                                          Battery (1): 97% [Charging, AC [redacted]]
```

```
ser-tech@Elieser-tech: ~/Id_Linux/Documents
~/Id_Linux/Documents

ser-tech@Elieser-tech:~/Id_Linux/Documents$ journalctl
Oct 28 18:07:49 localhost.localdomain kernel: Linux version 7.0.0-30-generic (buildd@lcy02-amd64-048) (x86_64-linux-gnu-gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0,
Oct 28 18:07:49 localhost.localdomain kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-7.0.0-30-generic root=UUID=8b234f9d-fcb3-4afa-bbba-1fd9a1e11c98 ro crash
Oct 28 18:07:49 localhost.localdomain kernel: KERNEL supported cpus:
Oct 28 18:07:49 localhost.localdomain kernel:   Intel GenuineIntel
Oct 28 18:07:49 localhost.localdomain kernel:   AMD AuthenticAMD
Oct 28 18:07:49 localhost.localdomain kernel:   Hygon HygonGenuine
Oct 28 18:07:49 localhost.localdomain kernel:   Centaur CentaurHauls
Oct 28 18:07:49 localhost.localdomain kernel:   zhaoxin   Shanghai
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-provided physical RAM map:
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [gap 0x000000000009a000-0x000000000009ffff]
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000bdeaefff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000bdeaf000-0x00000000bdffefff]  device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000bdfff000-0x00000000bdffffff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000be000000-0x00000000bfffffff]  device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000d0000000-0x00000000d0ffffff]  ACPI data
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000d1000000-0x00000000d1ffffff]  ACPI NVS
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000d2000000-0x00000000f1ffffff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000f2000000-0x00000000f20fffff]  device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000f2100000-0x00000000f21fffff]  ACPI NVS
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000f2200000-0x00000000f22fffff]  System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000f2300000-0x[redacted]]  device reserved
```

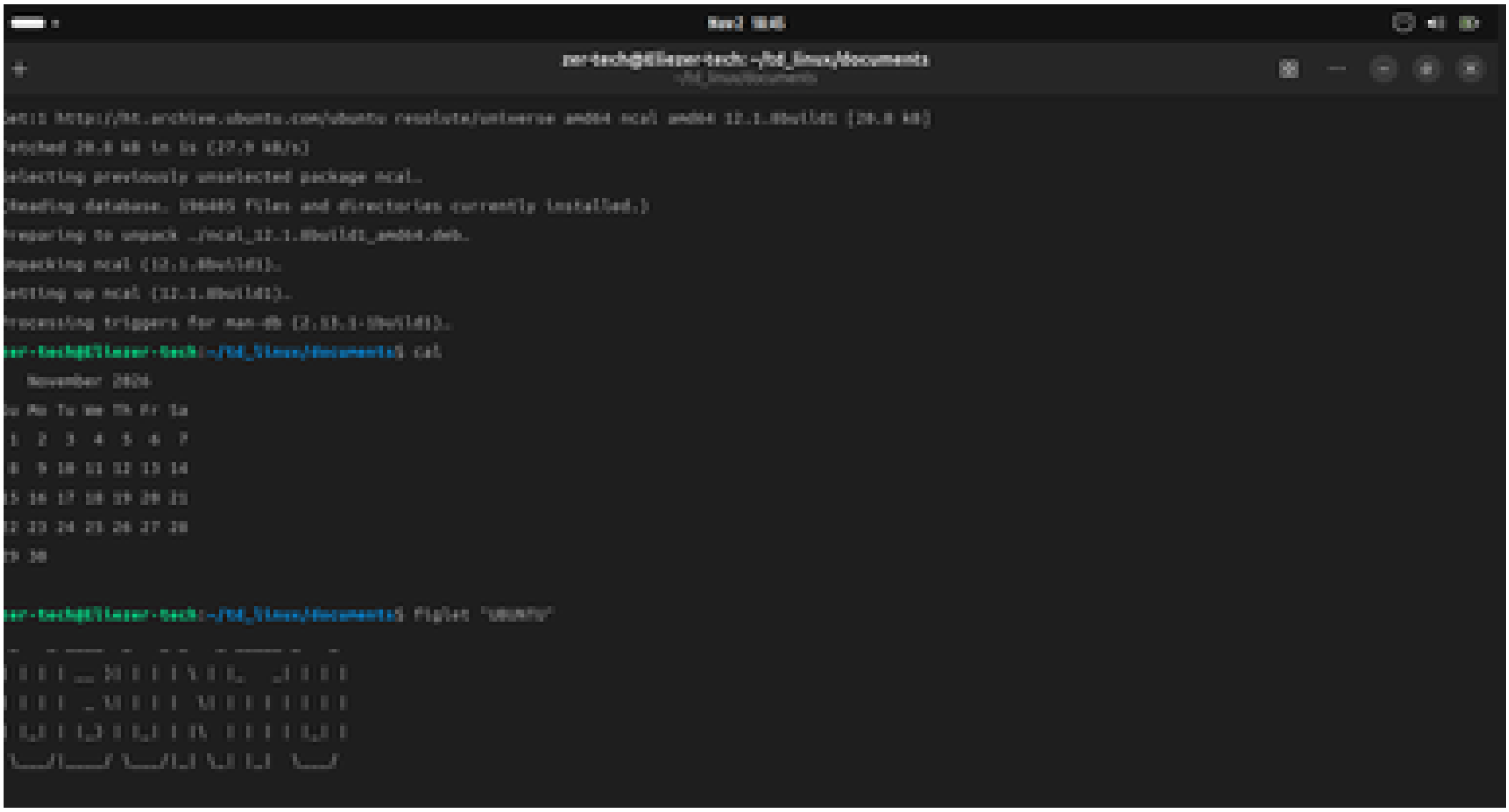

sudo dmesg
[sudo: authenticate] Password:
[0.000000] Linux version 7.0.0-42-generic (buildd@lcy02-amd64-049) (x86_64-linux-gnu-gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, GNU ld (GNU Binutils for Ubuntu) 2.46) #42-Ubuntu SMP PREEMPT_DYNAMIC Thu Oct 23 08:36:44 UTC 2026 (Ubuntu 7.0.0-42.42-generic 7.0.14)
[0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-7.0.0-42-generic root=UUID=0b126f9d-fcb1-4af4-bbba-1fd9a1e91c90 ro crashkernel=2G-4G:320M,4G-32G:512M,32G-64G:1024M,64G-128G:2048M,128G-:4096M
[0.000000] KERNEL supported cpus:
[0.000000] Intel GenuineIntel
[0.000000] AMD AuthenticAMD
[0.000000] Hygon HygonGenuine
[0.000000] Centaur CentaurHauls
[0.000000] zhaoxin Shanghai
[0.000000] x86/CPU: Model not found in latest microcode list
[0.000000] BIOS-provided physical RAM map:
[0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009fbff] System RAM
[0.000000] BIOS-e820: [gap 0x000000000009fc00-0x000000000009ffff]
[0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000bfeaefff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000bfeaf000-0x00000000bfffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000feffc000-0x00000000feffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fed00000-0x00000000fed03fff] ACPI data
[0.000000] BIOS-e820: [mem 0x00000000fed04000-0x00000000fed04fff] ACPI NVS
[0.000000] BIOS-e820: [mem 0x00000000fed05000-0x00000000fed1bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fed1c000-0x00000000fed1ffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fed20000-0x00000000fed3ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fed40000-0x00000000fed44fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fed45000-0x00000000fed8ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fed90000-0x00000000fed90fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fed91000-0x00000000fed91fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fed92000-0x00000000fed92fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fed93000-0x00000000fed93fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fed94000-0x00000000fed97fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fed98000-0x00000000fed98fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fed99000-0x00000000fed99fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fed9a000-0x00000000fed9afff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fed9b000-0x00000000fed9bfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fed9c000-0x00000000fed9ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000feda0000-0x00000000feda1fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000feda2000-0x00000000fedaffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fedb0000-0x00000000fedbffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fedc0000-0x00000000fedcffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fedd0000-0x00000000fedfffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] Local APIC
[0.000000] BIOS-e820: [mem 0x00000000fee01000-0x00000000fee01fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee02000-0x00000000fee02fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee03000-0x00000000fee03fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee04000-0x00000000fee04fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee05000-0x00000000fee05fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee06000-0x00000000fee06fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee07000-0x00000000fee07fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee08000-0x00000000fee08fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee09000-0x00000000fee09fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee0a000-0x00000000fee0afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee0b000-0x00000000fee0bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee0c000-0x00000000fee0cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee0d000-0x00000000fee0dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee0e000-0x00000000fee0efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee0f000-0x00000000fee0ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee10000-0x00000000fee10fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee11000-0x00000000fee11fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee12000-0x00000000fee12fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee13000-0x00000000fee13fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee14000-0x00000000fee14fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee15000-0x00000000fee15fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee16000-0x00000000fee16fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee17000-0x00000000fee17fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee18000-0x00000000fee18fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee19000-0x00000000fee19fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee1a000-0x00000000fee1afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee1b000-0x00000000fee1bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee1c000-0x00000000fee1cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee1d000-0x00000000fee1dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee1e000-0x00000000fee1efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee1f000-0x00000000fee1ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee20000-0x00000000fee20fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee21000-0x00000000fee21fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee22000-0x00000000fee22fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee23000-0x00000000fee23fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee24000-0x00000000fee24fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee25000-0x00000000fee25fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee26000-0x00000000fee26fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee27000-0x00000000fee27fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee28000-0x00000000fee28fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee29000-0x00000000fee29fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee2a000-0x00000000fee2afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee2b000-0x00000000fee2bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee2c000-0x00000000fee2cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee2d000-0x00000000fee2dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee2e000-0x00000000fee2efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee2f000-0x00000000fee2ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee30000-0x00000000fee30fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee31000-0x00000000fee31fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee32000-0x00000000fee32fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee33000-0x00000000fee33fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee34000-0x00000000fee34fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee35000-0x00000000fee35fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee36000-0x00000000fee36fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee37000-0x00000000fee37fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee38000-0x00000000fee38fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee39000-0x00000000fee39fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee3a000-0x00000000fee3afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee3b000-0x00000000fee3bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee3c000-0x00000000fee3cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee3d000-0x00000000fee3dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee3e000-0x00000000fee3efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee3f000-0x00000000fee3ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee40000-0x00000000fee40fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee41000-0x00000000fee41fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee42000-0x00000000fee42fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee43000-0x00000000fee43fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee44000-0x00000000fee44fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee45000-0x00000000fee45fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee46000-0x00000000fee46fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee47000-0x00000000fee47fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee48000-0x00000000fee48fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee49000-0x00000000fee49fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee4a000-0x00000000fee4afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee4b000-0x00000000fee4bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee4c000-0x00000000fee4cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee4d000-0x00000000fee4dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee4e000-0x00000000fee4efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee4f000-0x00000000fee4ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee50000-0x00000000fee50fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee51000-0x00000000fee51fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee52000-0x00000000fee52fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee53000-0x00000000fee53fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee54000-0x00000000fee54fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee55000-0x00000000fee55fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee56000-0x00000000fee56fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee57000-0x00000000fee57fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee58000-0x00000000fee58fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee59000-0x00000000fee59fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee5a000-0x00000000fee5afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee5b000-0x00000000fee5bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee5c000-0x00000000fee5cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee5d000-0x00000000fee5dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee5e000-0x00000000fee5efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee5f000-0x00000000fee5ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee60000-0x00000000fee60fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee61000-0x00000000fee61fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee62000-0x00000000fee62fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee63000-0x00000000fee63fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee64000-0x00000000fee64fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee65000-0x00000000fee65fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee66000-0x00000000fee66fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee67000-0x00000000fee67fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee68000-0x00000000fee68fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee69000-0x00000000fee69fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee6a000-0x00000000fee6afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee6b000-0x00000000fee6bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee6c000-0x00000000fee6cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee6d000-0x00000000fee6dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee6e000-0x00000000fee6efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee6f000-0x00000000fee6ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee70000-0x00000000fee70fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee71000-0x00000000fee71fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee72000-0x00000000fee72fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee73000-0x00000000fee73fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee74000-0x00000000fee74fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee75000-0x00000000fee75fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee76000-0x00000000fee76fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee77000-0x00000000fee77fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee78000-0x00000000fee78fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee79000-0x00000000fee79fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee7a000-0x00000000fee7afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee7b000-0x00000000fee7bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee7c000-0x00000000fee7cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee7d000-0x00000000fee7dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee7e000-0x00000000fee7efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee7f000-0x00000000fee7ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee80000-0x00000000fee80fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee81000-0x00000000fee81fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee82000-0x00000000fee82fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee83000-0x00000000fee83fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee84000-0x00000000fee84fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee85000-0x00000000fee85fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee86000-0x00000000fee86fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee87000-0x00000000fee87fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee88000-0x00000000fee88fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee89000-0x00000000fee89fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee8a000-0x00000000fee8afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee8b000-0x00000000fee8bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee8c000-0x00000000fee8cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee8d000-0x00000000fee8dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee8e000-0x00000000fee8efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee8f000-0x00000000fee8ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee90000-0x00000000fee90fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee91000-0x00000000fee91fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee92000-0x00000000fee92fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee93000-0x00000000fee93fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee94000-0x00000000fee94fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee95000-0x00000000fee95fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee96000-0x00000000fee96fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee97000-0x00000000fee97fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee98000-0x00000000fee98fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee99000-0x00000000fee99fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee9a000-0x00000000fee9afff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee9b000-0x00000000fee9bfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee9c000-0x00000000fee9cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee9d000-0x00000000fee9dfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000fee9e000-0x00000000fee9efff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000fee9f000-0x00000000fee9ffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000feea0000-0x00000000feea0fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000feea1000-0x00000000feea1fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000feea2000-0x00000000feeaffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000feeb0000-0x00000000feebffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000feec0000-0x00000000feecffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000feed0000-0x00000000feedffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000feee0000-0x00000000feefffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000ff000000-0x00000000ffffffff] reserved
[0.000000] BIOS-e820: [mem 0x0000000100000000-0x000000bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000000c000000000-0x000000ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000010000000000-0x0000013fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000014000000000-0x0000017fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000018000000000-0x000001bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000001c000000000-0x000001ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000020000000000-0x0000023fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000024000000000-0x0000027fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000028000000000-0x000002bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000002c000000000-0x000002ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000030000000000-0x0000033fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000034000000000-0x0000037fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000038000000000-0x000003bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000003c000000000-0x000003ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000040000000000-0x0000043fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000044000000000-0x0000047fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000048000000000-0x000004bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000004c000000000-0x000004ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000050000000000-0x0000053fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000054000000000-0x0000057fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000058000000000-0x000005bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000005c000000000-0x000005ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000060000000000-0x0000063fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000064000000000-0x0000067fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000068000000000-0x000006bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000006c000000000-0x000006ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000070000000000-0x0000073fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000074000000000-0x0000077fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000078000000000-0x000007bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000007c000000000-0x000007ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000080000000000-0x0000083fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000084000000000-0x0000087fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000088000000000-0x000008bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000008c000000000-0x000008ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000090000000000-0x0000093fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x0000094000000000-0x0000097fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x0000098000000000-0x000009bfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x000009c000000000-0x000009ffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000a0000000000-0x00000a3fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000a4000000000-0x00000a7fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000a8000000000-0x00000abfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000ac000000000-0x00000affffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000b0000000000-0x00000b3fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000b4000000000-0x00000b7fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000b8000000000-0x00000bbfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000bc000000000-0x00000bffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000c0000000000-0x00000c3fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000c4000000000-0x00000c7fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000c8000000000-0x00000cbfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000cc000000000-0x00000cffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000d0000000000-0x00000d3fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000d4000000000-0x00000d7fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000d8000000000-0x00000dbfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000dc000000000-0x00000dffffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000e0000000000-0x00000e3fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000e4000000000-0x00000e7fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000e8000000000-0x00000ebfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000ec000000000-0x00000effffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000f0000000000-0x00000f3fffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000f4000000000-0x00000f7fffffffff] device reserved
[0.000000] BIOS-e820: [mem 0x00000f8000000000-0x00000fbfffffffff] System RAM
[0.000000] BIOS-e820: [mem 0x00000fc000000000-0x00000fffffffffff] device reserved
[0.000000] NX (Execute Disable) protection: active
[0.000000] APIC: Static calls initialized
[0.000000] SMBIOS 3.0 present.
[0.000000] DMI: Dell Inc. OptiPlex 7040/0HG4P4, BIOS 1.4.7 08/28/2015
[0.000000] DMI: Memory slots populated: 2/4
[0.000000] Hypervisor detected: KVM
[0.000000] kvm-clock: Using msrs 4b564d01 and 4b564d00
[0.000000] kvm-clock: using sched offset of 4 cycles
[0.000000] clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns
[0.000000] tsc: Detected 3400.000 MHz processor
[0.000000] tsc: Detected 3399.996 MHz TSC
[0.000000] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[0.000000] e820: remove [mem 0x000a0000-0x000fffff] usable
[0.000000] last_pfn = 0x140000 max_arch_pfn = 0x400000000
[0.000000] MTRR map: 4 entries (3 fixed + 1 variable; max 19), built from 8 variable MTRRs
[0.000000] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
[0.000000] last_pfn = 0xbfeaf max_arch_pfn = 0x400000000
[0.000000] Using GB pages for direct mapping
[0.000000] Secure boot disabled
[0.000000] RAMDISK: [mem 0x3a9ca000-0x3fffffff]
[0.000000] ACPI: Early table checksum verification disabled
[0.000000] ACPI: RSDP 0x00000000000F0490 000024 (v02 DELL)
[0.000000] ACPI: XSDT 0x00000000BFEB1188 0000AC (v01 DELL CBX3 01072009 AMI 00010013)
[0.000000] ACPI: FACP 0x00000000BFEB0000 0000F4 (v03 DELL CBX3 01072009 AMI 00010013)
[0.000000] ACPI: DSDT 0x00000000BFEB0290 00A1C1 (v02 DELL CBX3 01072009 INTL 20091112)
[0.000000] ACPI: FACS 0x00000000BFEC0000 000040
[0.000000] ACPI: UEFI 0x00000000BFEB0240 000042 (v01 DELL CBX3 01072009 AMI 00010013)
[0.000000] ACPI: SSDT 0x00000000BFEB0200 000040 (v01 DELL CBX3 01072009 AMI 00010013)
[0.000000] ACPI: MCFG 0x00000000BFEB0150 0000AC (v01 DELL CBX3 01072009 MSFT 00000097)
[0.000000] ACPI: HPET 0x00000000BFEB0110 000038 (v01 DELL CBX3 01072009 AMI. 00000005)
[0.000000] ACPI: APIC 0x00000000BFEB0A50 000084 (v03 DELL CBX3 01072009 AMI 00010013)
[0.000000] ACPI: SSDT 0x00000000BFEB0AE0 000144 (v02 SataRe SataTbl 00001000 INTL 20091112)
[0.000000] ACPI: SSDT 0x00000000BFEB0C30 0000DE (v02 SataRe SataAhci 00001000 INTL 20091112)
[0.000000] ACPI: SSDT 0x00000000BFEB0D10 000317 (v02 DELL CBX3 01072009 INTL 20091112)
[0.000000] ACPI: SSDT 0x00000000BFEB1030 000056 (v02 DELL CBX3 01072009 INTL 20091112)
[0.000000] ACPI: SSDT 0x00000000BFEB1090 000042 (v02 DELL CBX3 01072009 INTL 20091112)
[0.000000] ACPI: Reserving FACP table memory at [mem 0xbfeb0000-0xbfeb00f3]
[0.000000] ACPI: Reserving DSDT table memory at [mem 0xbfeb0290-0xbfeba450]
[0.000000] ACPI: Reserving FACS table memory at [mem 0xbfec0000-0xbfec003f]
[0.000000] ACPI: Reserving UEFI table memory at [mem 0xbfeb0240-0xbfeb0281]
[0.000000] ACPI: Reserving SSDT table memory at [mem 0xbfeb0200-0xbfeb023f]
[0.000000] ACPI: Reserving MCFG table memory at [mem 0xbfeb0150-0xbfeb01fb]
[0.000000] ACPI: Reserving HPET table memory at [mem 0xbfeb0110-0xbfeb0147]
[0.000000] ACPI: Reserving APIC table memory at [mem 0xbfeb0a50-0xbfeb0ad3]
[0.000000] ACPI: Reserving SSDT table memory at [mem 0xbfeb0ae0-0xbfeb0c23]
[0.000000] ACPI: Reserving SSDT table memory at [mem 0xbfeb0c30-0xbfeb0d0d]
[0.000000] ACPI: Reserving SSDT table memory at [mem 0xbfeb0d10-0xbfeb1026]
[0.000000] ACPI: Reserving SSDT table memory at [mem 0xbfeb1030-0xbfeb1085]
[0.000000] ACPI: Reserving SSDT table memory at [mem 0xbfeb1090-0xbfeb10d1]
[0.000000] No NUMA configuration found
[0.000000] Faking a node at [mem 0x0000000000000000-0x000000140000ffff]
[0.000000] NODE_DATA(0) allocated [mem 0x140000c0c0-0x140000ffff]
[0.000000] Zone ranges:
[0.000000] DMA [mem 0x0000000000001000-0x0000000000ffffff]
[0.000000] DMA32 [mem 0x0000000001000000-0x00000000ffffffff]
[0.000000] Normal [mem 0x0000000100000000-0x000000140000ffff]
[0.000000] Device empty
[0.000000] Movable zone start for each node
[0.000000] Early memory node ranges
[0.000000] node 0: [mem 0x0000000000001000-0x000000000009efff]
[0.000000] node 0: [mem 0x0000000000100000-0x00000000bfeaefff]
[0.000000] node 0: [mem 0x00000000feffc000-0x00000000feffffff]
[0.000000] node 0: [mem 0x00000000fed05000-0x00000000fed1bfff]
[0.000000] node 0: [mem 0x00000000fed20000-0x00000000fed3ffff]
[0.000000] node 0: [mem 0x00000000fed45000-0x00000000fed8ffff]
[0.000000] node 0: [mem 0x00000000fed91000-0x00000000fed91fff]
[0.000000] node 0: [mem 0x00000000fed93000-0x00000000fed93fff]
[0.000000] node 0: [mem 0x00000000fed98000-0x00000000fed98fff]
[0.000000] node 0: [mem 0x00000000fed9a000-0x00000000fed9afff]
[0.000000] node 0: [mem 0x00000000fed9c000-0x00000000fed9ffff]
[0.000000] node 0: [mem 0x00000000feda2000-0x00000000fedaffff]
[0.000000] node 0: [mem 0x00000000fedc0000-0x00000000fedcffff]
[0.000000] node 0: [mem 0x00000000fee01000-0x00000000fee01fff]
[0.000000] node 0: [mem 0x00000000fee03000-0x00000000fee03fff]
[0.000000] node 0: [mem 0x00000000fee05000-0x00000000fee05fff]
[0.000000] node 0: [mem 0x00000000fee07000-0x00000000fee07fff]
[0.000000] node 0: [mem 0x00000000fee09000-0x00000000fee09fff]
[0.000000] node 0: [mem 0x00000000fee0b000-0x00000000fee0bfff]
[0.000000] node 0: [mem 0x00000000fee0d000-0x00000000fee0dfff]
[0.000000] node 0: [mem 0x00000000fee0f000-0x00000000fee0ffff]
[0.000000] node 0: [mem 0x00000000fee11000-0x00000000fee11fff]
[0.000000] node 0: [mem 0x00000000fee13000-0x00000000fee13fff]
[0.000000] node 0: [mem 0x00000000fee15000-0x00000000fee15fff]
[0.000000] node 0: [mem 0x00000000fee17000-0x00000000fee17fff]
[0.000000] node 0: [mem 0x00000000fee19000-0x00000000fee19fff]
[0.000000] node 0: [mem 0x00000000fee1b000-0x00000000fee1bfff]

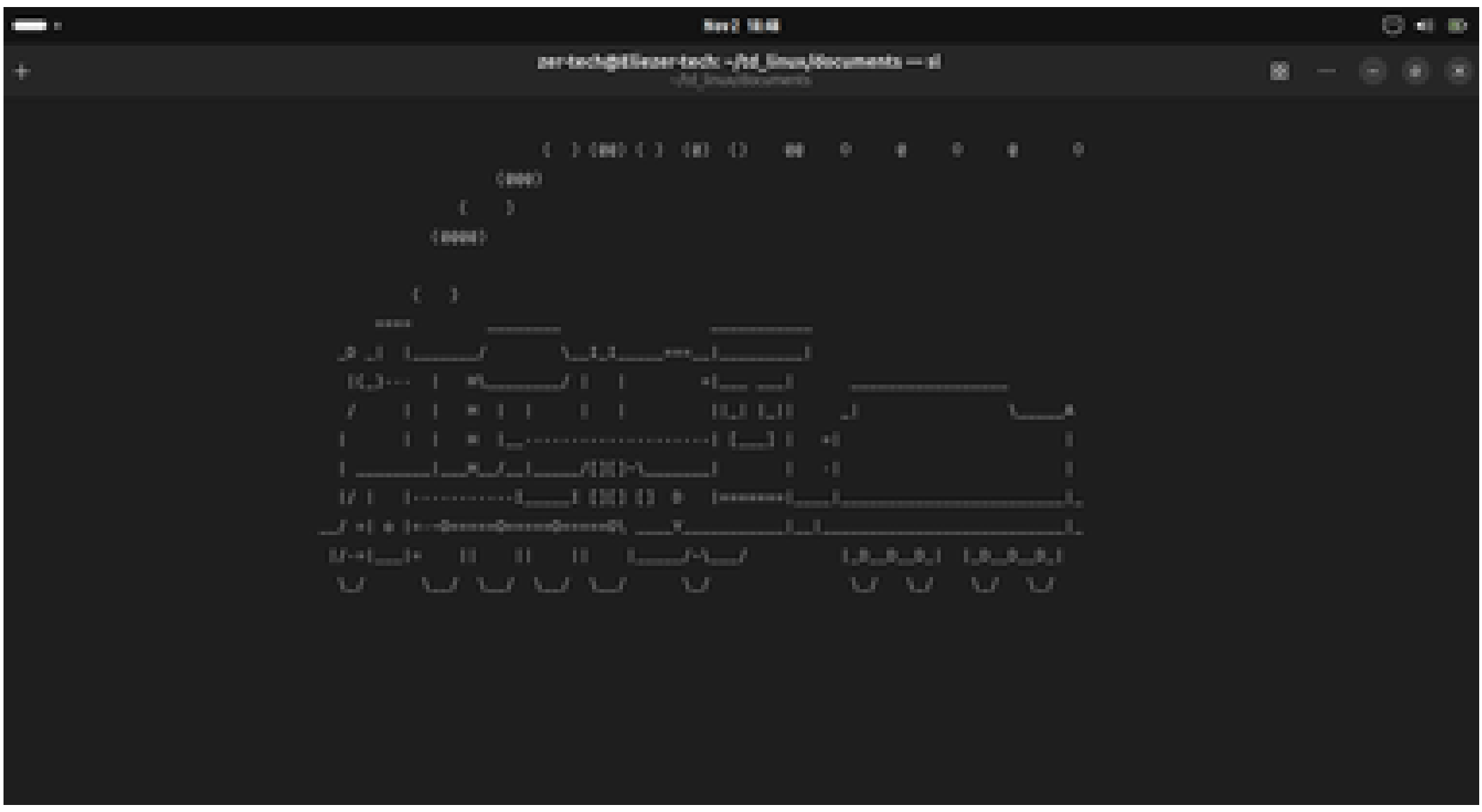
```
[  153.888478] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 4 times, consider switching to wq_UNBOUND
zer-tech@ElAzar-tech:~/Pd_linux/documents$ sudo dmesg | grep -i memory
[    0.000000] DMI: Memory slots populated: 0/0
[    0.163541] ACPI: Reserving FACP table memory at [mem 0xded79000-0xded794f1]
[    0.163546] ACPI: Reserving DSDT table memory at [mem 0xded7a000-0xded7c353]
[    0.163547] ACPI: Reserving FACS table memory at [mem 0xded7e000-0xded7e03f]
[    0.163548] ACPI: Reserving APIC table memory at [mem 0xded79000-0xded79059]
[    0.163549] ACPI: Reserving SSDT table memory at [mem 0xded77000-0xded77944]
[    0.163549] ACPI: Reserving BGRT table memory at [mem 0xded76000-0xded76037]
[    0.168501] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[    0.168567] PM: hibernation: Registered nosave memory: [mem 0x000a0000-0x000fffff]
[    0.168569] PM: hibernation: Registered nosave memory: [mem 0xdd3d3000-0xdd3f5fff]
[    0.168570] PM: hibernation: Registered nosave memory: [mem 0xdd6a3000-0xddf9afff]
[    0.168571] PM: hibernation: Registered nosave memory: [mem 0xdeae0000-0xdedfefff]
[    0.168572] PM: hibernation: Registered nosave memory: [mem 0xdf120000-0xdf132fff]
[    0.168574] PM: hibernation: Registered nosave memory: [mem 0xdf16e000-0xffffffff]
[    0.168663] Early memory node ranges
[    0.263457] Freeing SMP alternatives memory: 56K
[    0.379276] Memory: 15295488K/16574716K available (129004K kernel code, 4961K rwdata, 16739K rodata, 5552K init, 5032K bss, 1615484K reserved, 0K cma-reserved
)
[    0.381504] x86/mm: Memory block size: 128MB
[    0.389164] efi: Freeing EFI boot services memory: 2132K
[    1.045741] Freeing initrd memory: 57476K
[    1.130867] Freeing unused decrypted memory: 2028K
[    1.150899] Freeing unused kernel image (initmem) memory: 5552K
[    1.155499] Freeing unused kernel image (text/rodata gap) memory: 668K
```

```
 ncal

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 20.0 kB
  Space needed: 99.4 kB / 16.0 GB available

Get:1 http://ftl.archive.ubuntu.com/ubuntu resolute/universe amd64 ncal amd64 12.1.8build1 [20.0 kB]
Fetched 20.0 kB in 1s (27.9 kB/s)
Selecting previously unselected package ncal.
(Reading database... 196403 files and directories currently installed.)
Preparing to unpack .../ncal_12.1.8build1_amd64.deb...
Unpacking ncal (12.1.8build1)...
Setting up ncal (12.1.8build1)...
Processing triggers for man-db (2.13.1-1build1)...
zer-tech@ElAzar-tech:~/Pd_linux/documents$ cal
   November 2026
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30
```


Get:1 http://bt.archive.ubuntu.com/ubuntu resolute/universe amd64 ncal amd64 12.1.8build1 [20.8 kB]
Fetched 20.8 kB in 1s (27.9 kB/s)
Selecting previously unselected package ncal.
(Reading database ... 370405 files and directories currently installed.)
Preparing to unpack .../ncal_12.1.8build1_amd64.deb ...
Unpacking ncal (12.1.8build1) ...
Setting up ncal (12.1.8build1) ...
Processing triggers for man-db (2.13.1-1build1) ...
November 2026
Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30



```
user-tech@Linux-tech:~/Id_Linux/Documents$ ll
user-tech@Linux-tech:~/Id_Linux/Documents$ seq 1 10
1
2
3
4
5
6
7
8
9
10
user-tech@Linux-tech:~/Id_Linux/Documents$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
```

```
user-tech@Linux-tech:~/Id_Linux/Documents$ seq 1 10
1
2
3
4
5
6
7
8
9
10
user-tech@Linux-tech:~/Id_Linux/Documents$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
user-tech@Linux-tech:~/Id_Linux/Documents$ shuf -i 1-100 -n 1
12
user-tech@Linux-tech:~/Id_Linux/Documents$
```

EXERCICE 5

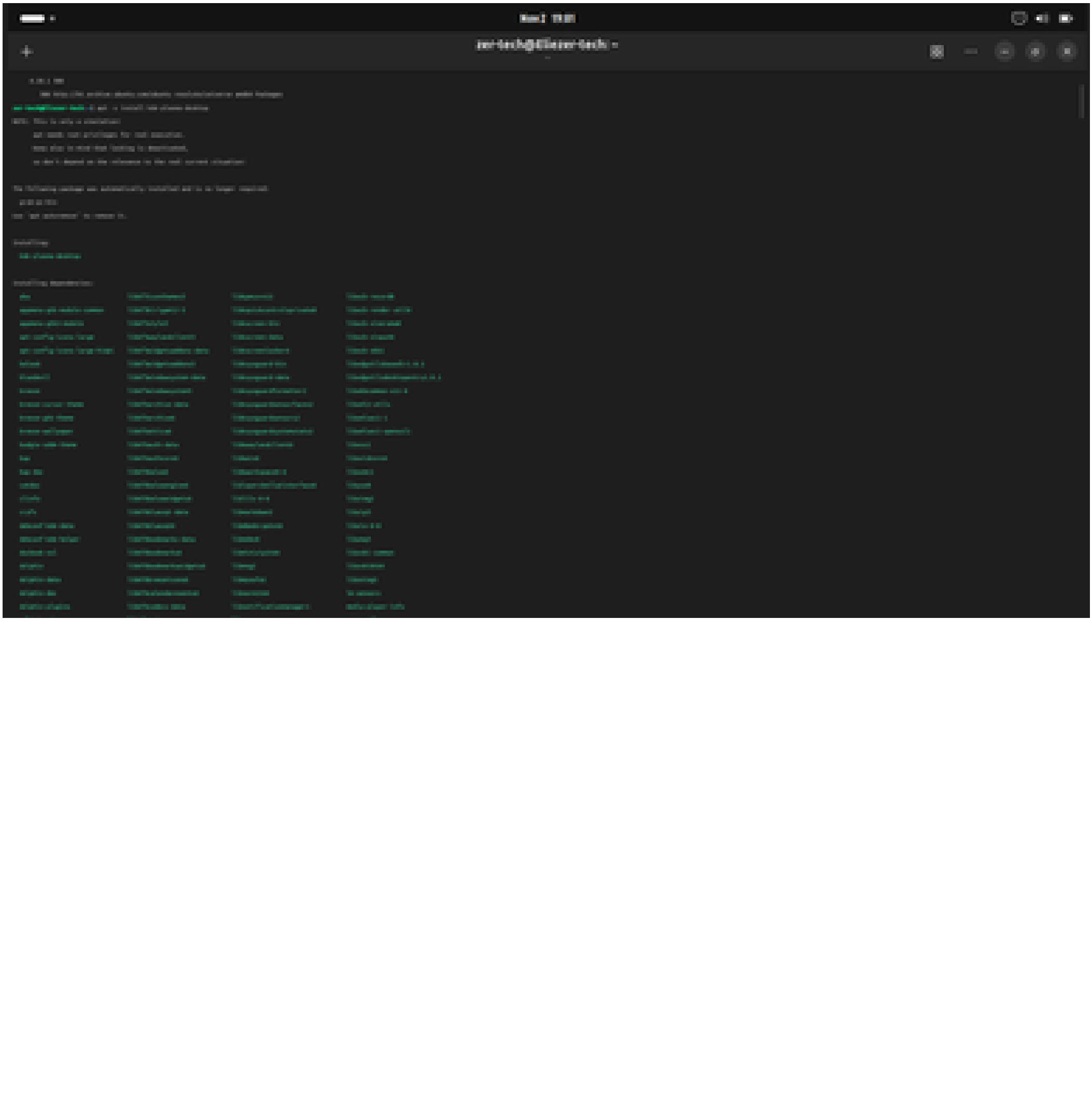
Cette expérience vise à installer, configurer et comparer deux environnements de bureau distincts (KDE Plasma et XFCE) afin d'analyser leur comportement et leur impact sur le système.

1. Identification de l'environnement de bureau actif et du type de session graphique (serveur d'affichage) via l'analyse des variables d'environnement du système.
2. Vérification de la disponibilité et du statut des paquets logiciels requis dans le gestionnaire de paquets de la distribution.
3. Simulation des installations pour évaluer en amont l'espace disque nécessaire et examiner l'arbre des dépendances logicielles de chaque interface.
4. Installation effective des deux environnements avec les privilèges d'administration, en prenant soin de conserver le gestionnaire de connexion d'origine en cas d'invite système.
5. Déconnexion de la session courante après sauvegarde des travaux, puis tests successifs d'ouverture de session sous Plasma, puis sous XFCE.

Évaluation et comparaison

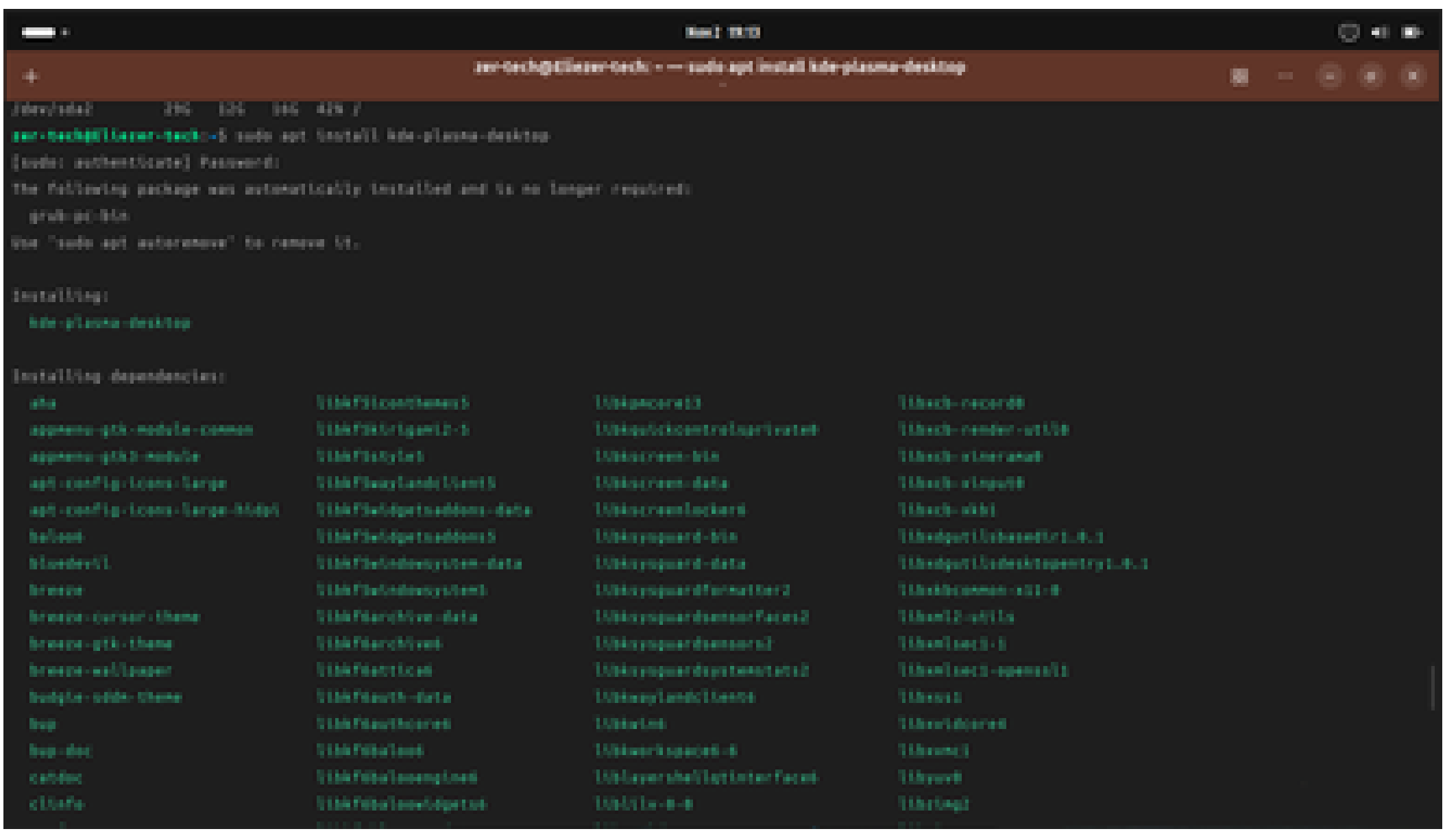
Pour chaque interface, un banc d'essai standardisé est appliqué afin de tester l'ergonomie et la configuration :

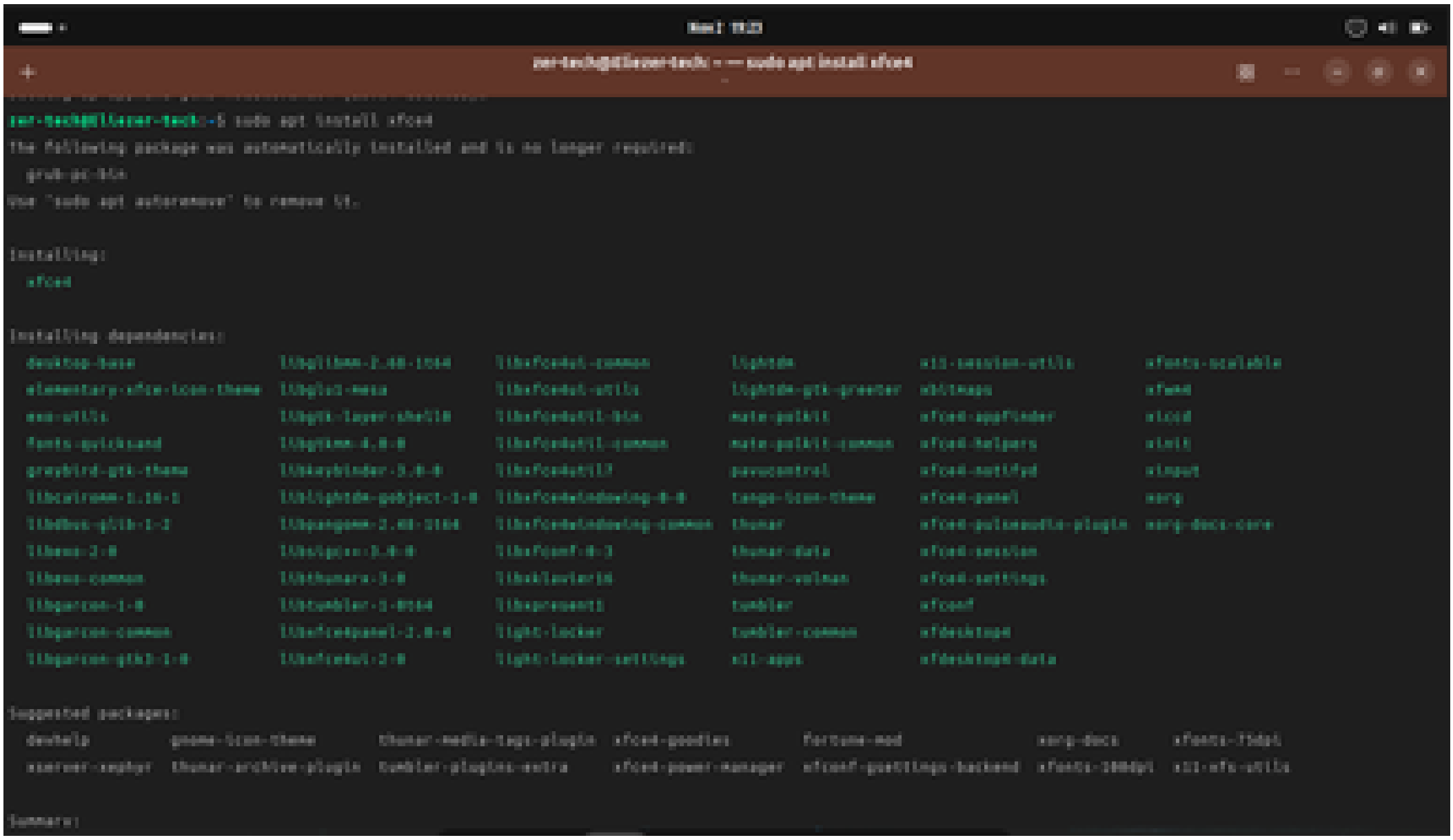
- L'exécution des outils de base (terminal, gestionnaire de fichiers).
- L'accès aux documents de travail.
- La personnalisation de l'interface (fond d'écran, création de raccourcis).
- L'exploration du panneau d'administration des paramètres système.



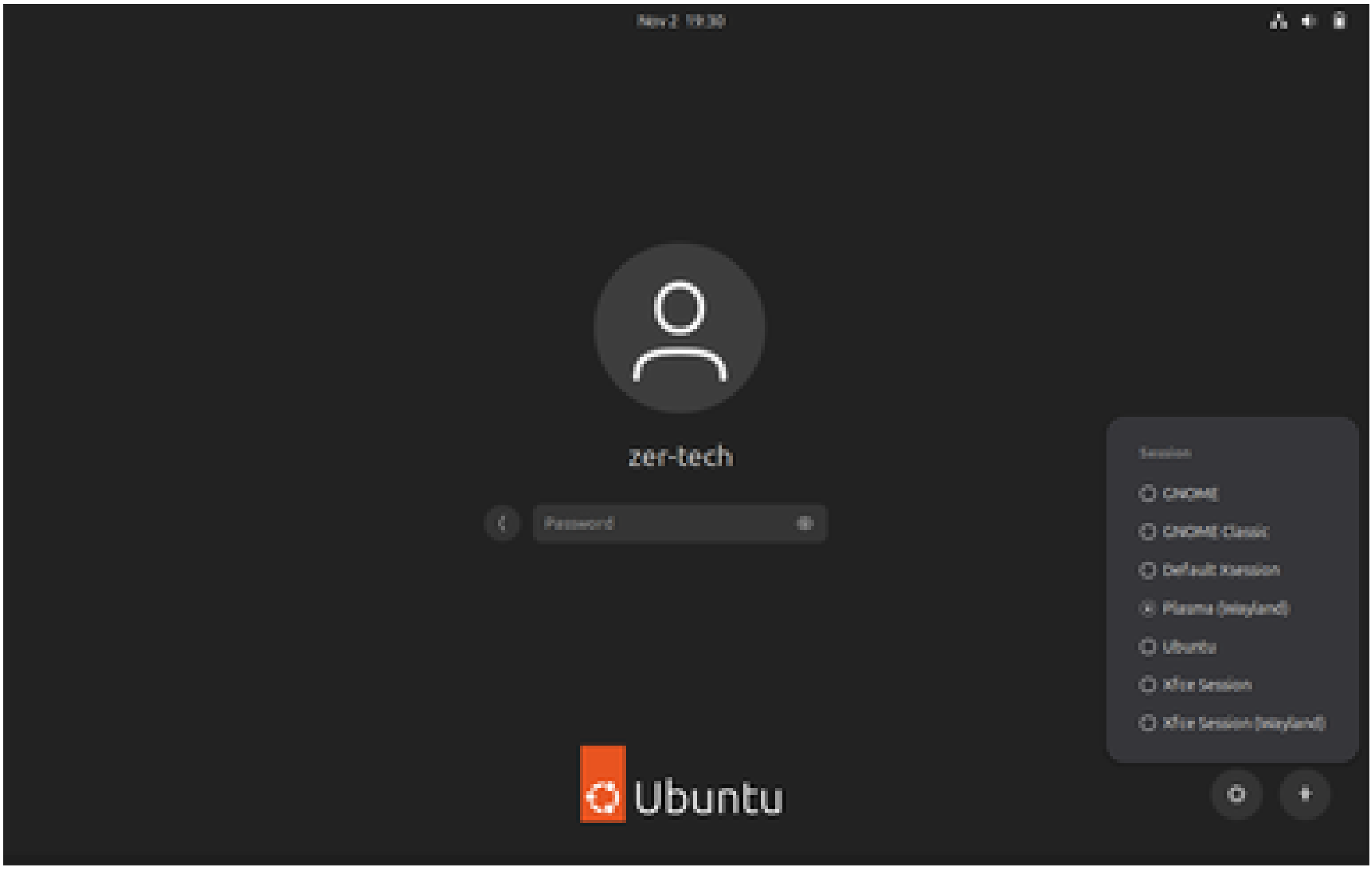


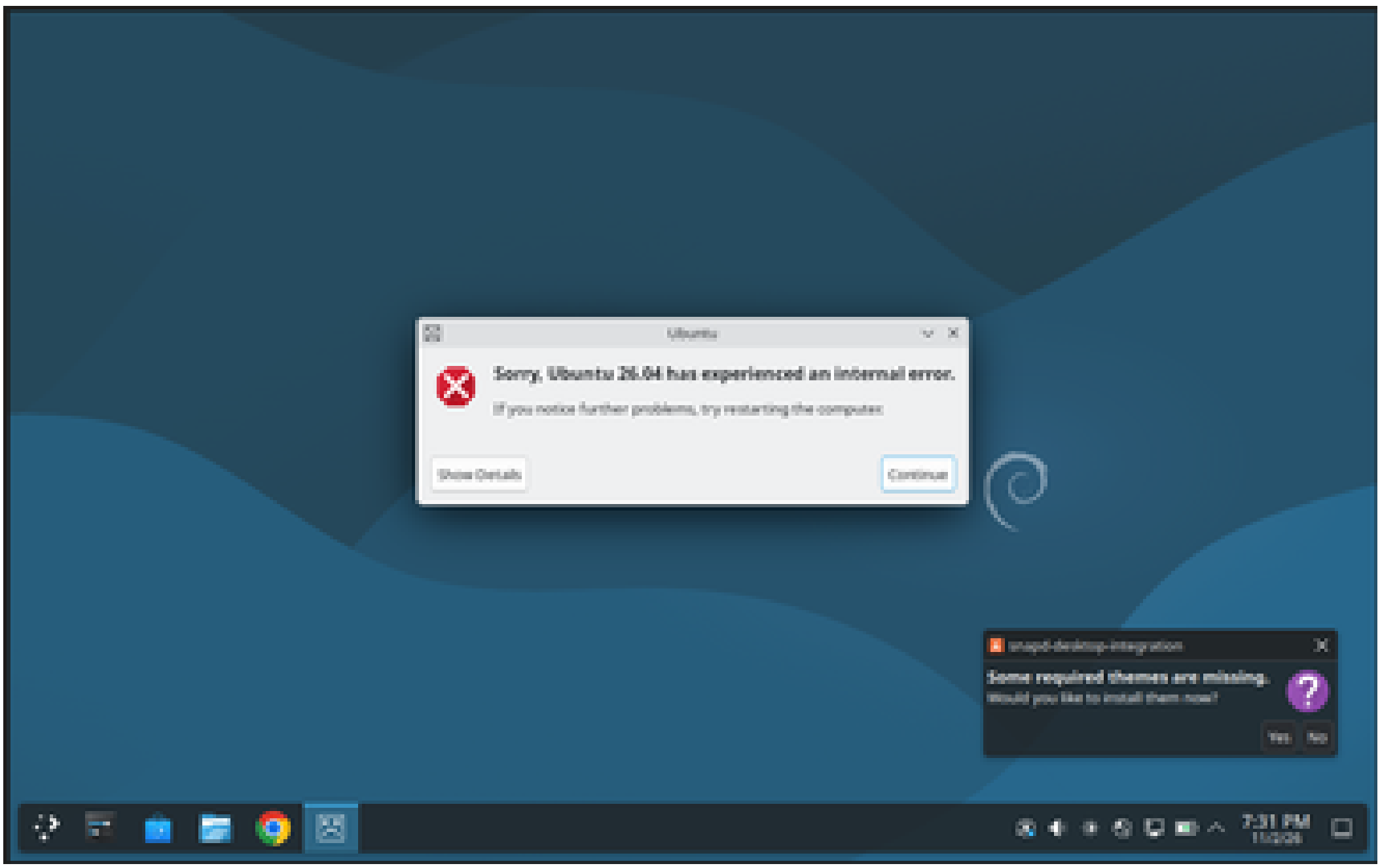

avr-tech@ElseAvr-tech:~$ df -h /
Filesystem   Size Used Avail Use% Mounted on
/dev/sda2   29G   12G   16G 42% /
avr-tech@ElseAvr-tech:~$

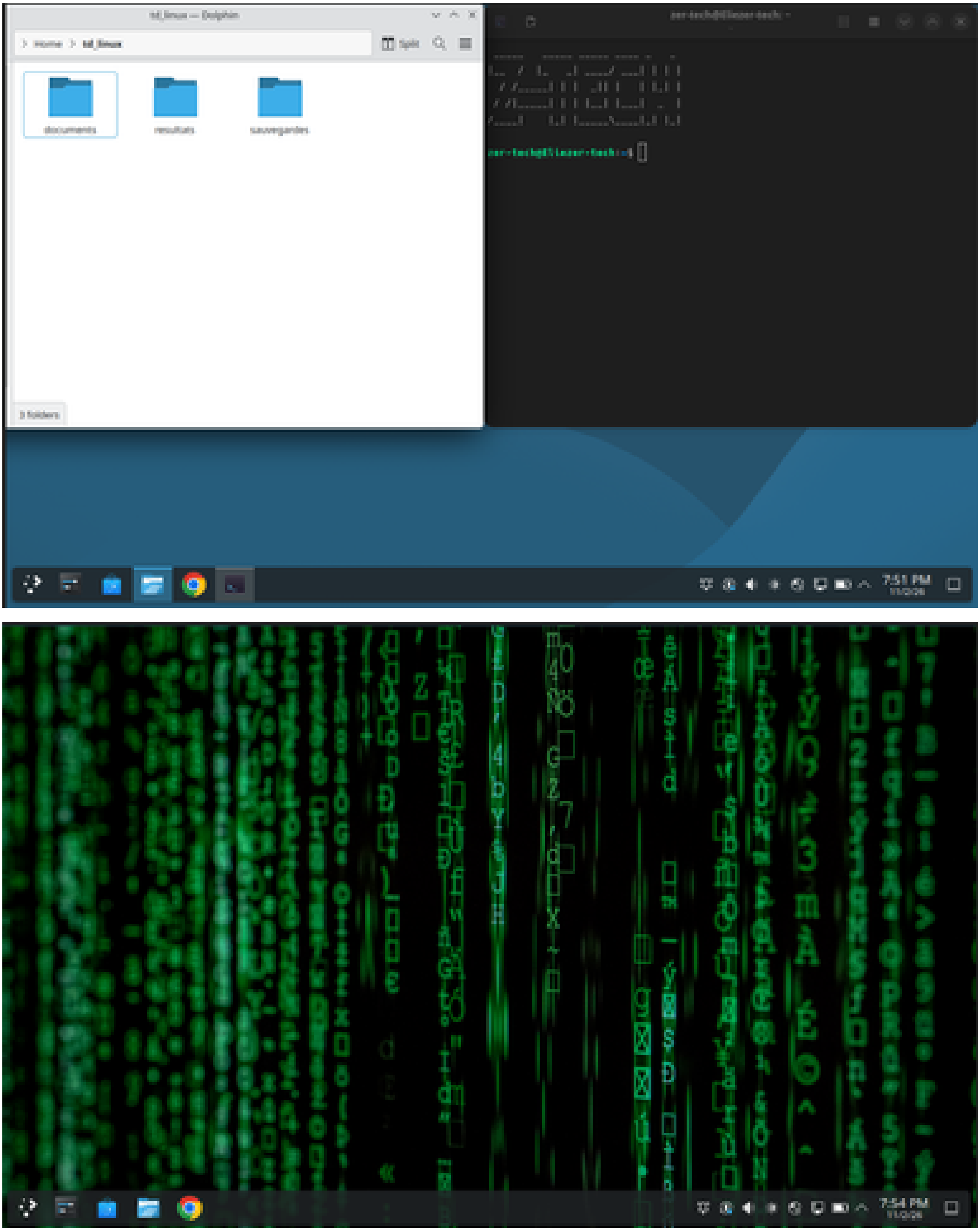

avr-tech@ElseAvr-tech: ~ — sudo apt install kde-plasma-desktop
/dev/sda2   29G   12G   16G 42% /
avr-tech@ElseAvr-tech:~$ sudo apt install kde-plasma-desktop
[sudo: authenticate] Password:
The following package was automatically installed and is no longer required:
  grub-pc-bin
Use 'sudo apt autoremove' to remove it.
Installing:
  kde-plasma-desktop
Installing dependencies:
  ark   libkf5iconthemes5   libqmacaroni3   libtech-recordd
  appmenu-gtk-module-common   libkf5kirigami2-5   libqqu5sicentralsprivate4 libtech-render-util0
  appmenu-gtk3-module   libkf5kyint   libkosraven-bin   libtech-vimeramu4
  apt-config-icons-large   libkf5waylandqtclient5   libkosraven-data   libtech-vimpu00
  apt-config-icons-large-hidpi libkf5webkitpatchaddons-data libkosravenlaqikard   libtech-vibj
  baloom4   libkf5webkitpatchaddons5   libkzysquard-bin   libxdgutilsbasedir1.0.1
  bluedevil   libkf5webkitsystem-data   libkzysquard-data   libxdgutilsdesktopentry1.0.1
  breeze   libkf5webkitsystem4   libkzysquardformatter2   libxkbcommon-x11-0
  breeze-cursor-theme   libkfbarchive-data   libkzysquardsensorfaces2   libxml2-utils
  breeze-gtk-theme   libkfbarchive4   libkzysquardsensors2   libxmlsec1-1
  breeze-wallpaper   libkfbattical4   libkzysquardsystemstats2   libxmlsec1-openssl1
  buddie-sddm-theme   libkfbauth-data   libkwaylandclient4   libxss1
  bup   libkfbauthcore4   libkwlm4   libxvidcore4
  bup-doc   libkfbbalen4   libkworkspace5-4   libxxhash0
  cantor   libkfbbalenengine4   liblayershellqtinterface4 libyuv0
  clinfo   libkfbbalenwidgets4   liblzlw-0-0   libzing1

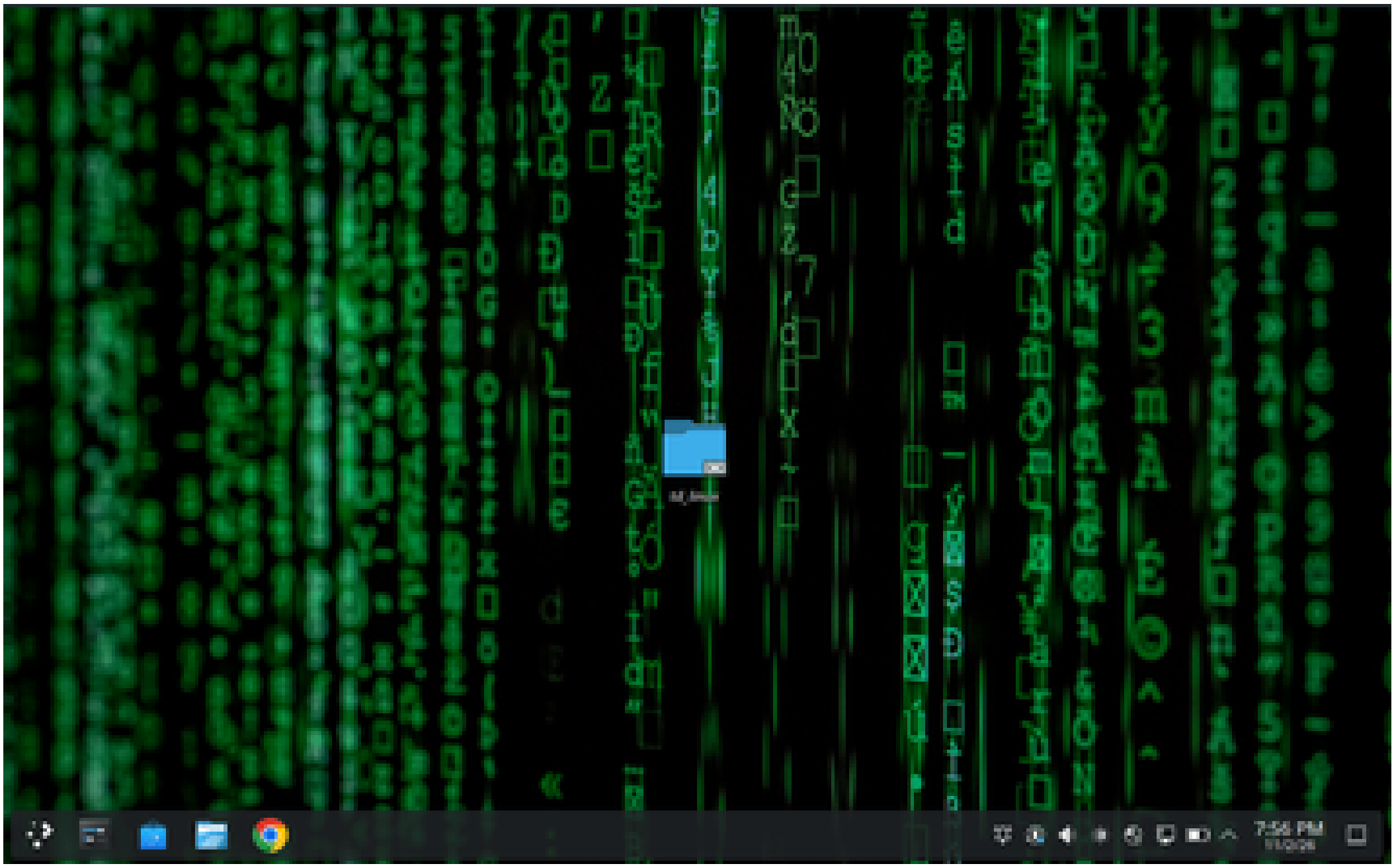

zer-tech@tilezer-tech: ~ — sudo apt install xfce4
zer-tech@tilezer-tech:~$ sudo apt install xfce4
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'sudo apt autoremove' to remove it.
Installing:
xfce4
Installing dependencies:
desktop-base
elementary-xfce-icon-theme
exo-utils
fonts-quicksand
greybird-gtk-theme
libcairomm-1.16-1
libdbus-glib-1-2
libexo-2-0
libexo-common
libgarcon-1-0
libgarcon-common
libgarcon-gtk3-1-0
libgdkmm-2.48-1544
libgio3-perl
libgtk3-layer-shell0
libgtkmm-4.0-0
libkeybinder-3.0-0
libpangomm-2.48-1164
libsigc++-3.0-0
libthunarx-3-0
libtumbler-1-0164
libwnck-3-0
libxfce4panel-2.0-4
libxfce4ui-2-0
libxfce4util-bin
libxfce4util-common
libxfce4util7
libxfce4windowing-0-0
libxfce4windowing-common
libxfconf-0-3
libxklavier16
libxpresent1
light-locker
light-locker-settings
lightdm
mate-polkit
mate-polkit-common
menucentral
tango-icon-theme
thunar
thunar-data
thunar-volman
tumbler
tumbler-common
x11-apps
xfce4-appfinder
xfce4-helpers
xfce4-notifyd
xfce4-panel
xfce4-pulseaudio-plugin
xfce4-session
xfce4-settings
xfconf
xfdesktop4
xfdesktop4-data
xfonts-scalable
xfwm4
xiccd
xinit
xinput
xorg
xorg-docs-core
Suggested packages:
devhelp
gnome-icon-theme
thunar-media-tags-plugin
xfce4-goodies
fortune-mod
xorg-docs
xfonts-75dpi
xserver-xephyr
thunar-archive-plugin
tumbler-plugins-extra
xfce4-power-manager
xfconf-gsettings-backend
xfonts-100dpi
x11-xfs-utils
Summary:

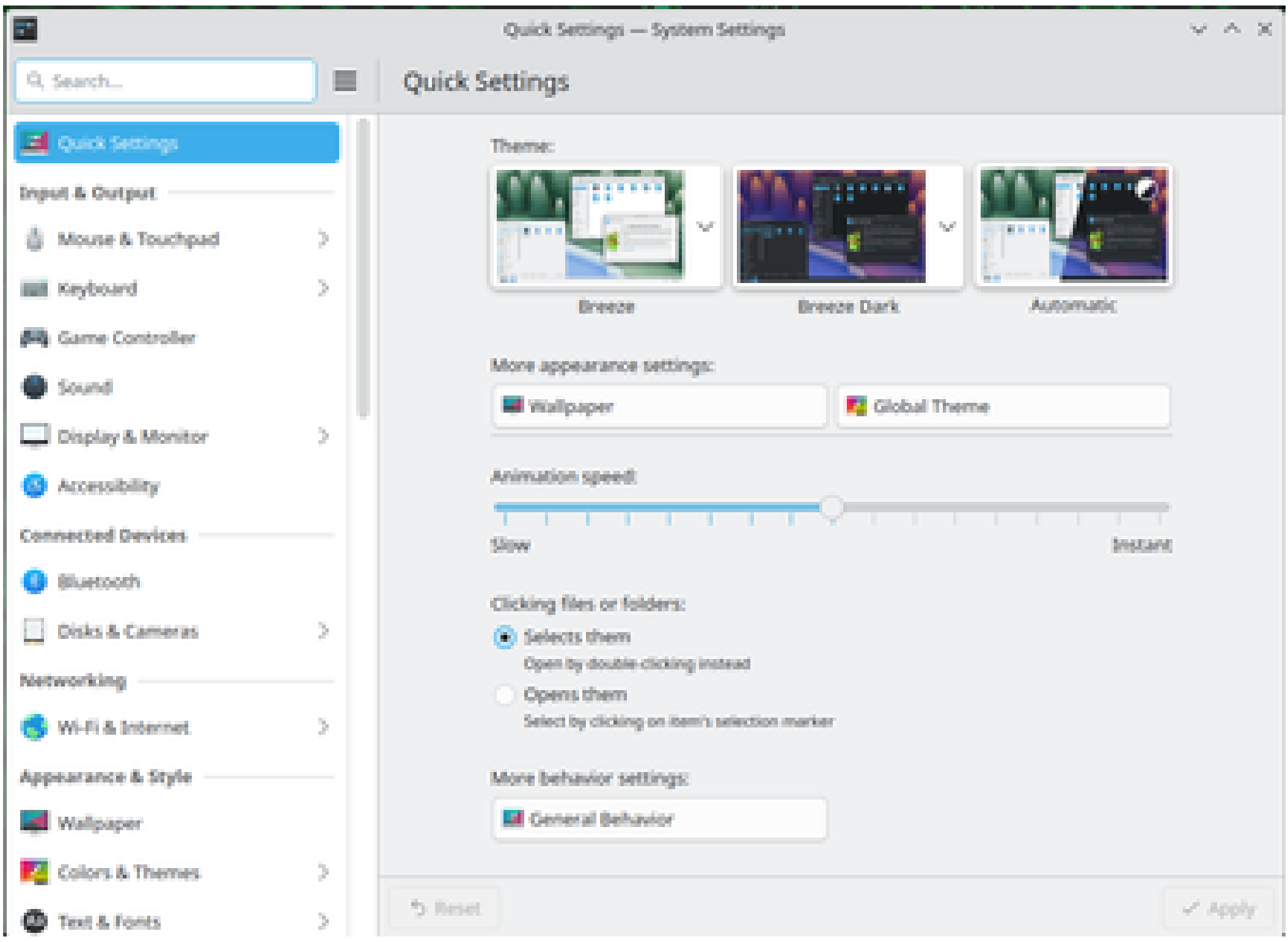

zer-tech@tilezer-tech: ~ — sudo apt install xfce4
Package configuration
Configuring lightdm
A display manager is a program that provides graphical login capabilities for the X Window System.
Only one display manager can manage a given X server, but multiple display manager packages are installed. Please select which display manager should run by default.
Multiple display managers can run simultaneously if they are configured to manage different servers; to achieve this, configure the display managers accordingly, edit each of their init scripts in /etc/init.d, and disable the check for a default display manager.
Default display manager:
gdm3
lightdm
sddm

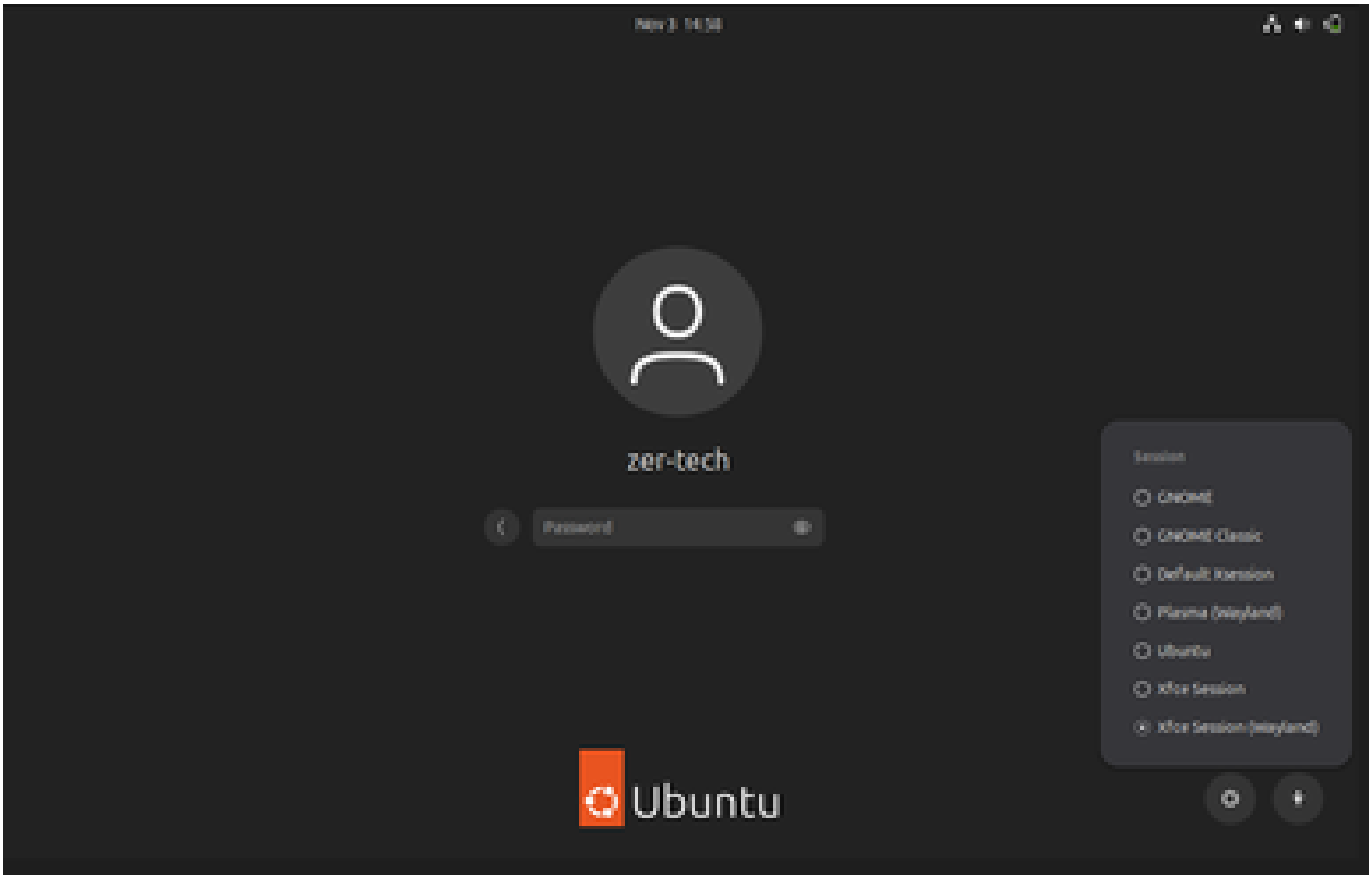

zer-tech
Password
Ubuntu

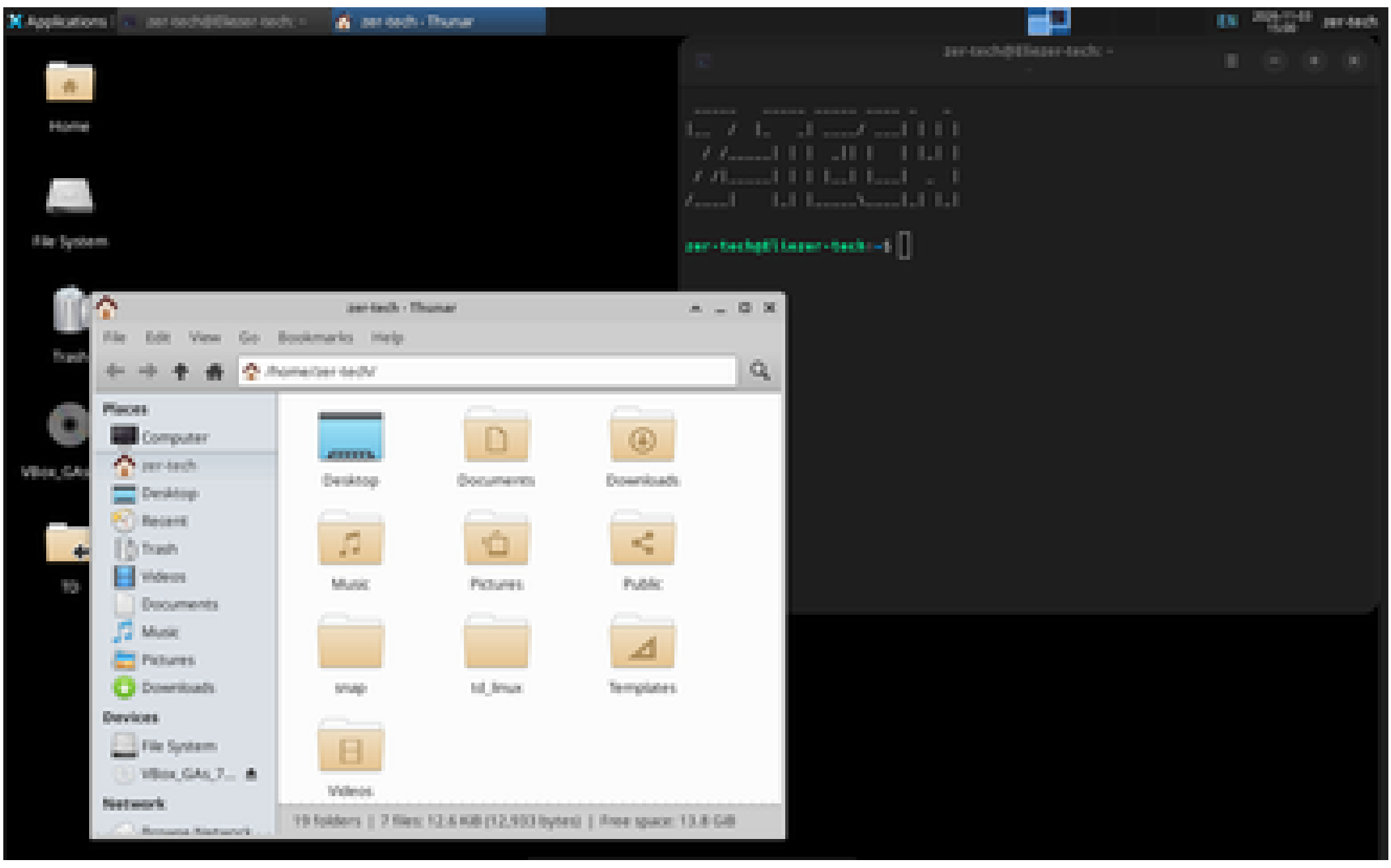

Ubuntu
Sorry, Ubuntu 26.04 has experienced an internal error.
If you notice further problems, try restarting the computer.
Show Details
Continue
Some required themes are missing.
Would you like to install them now?
7:31 PM

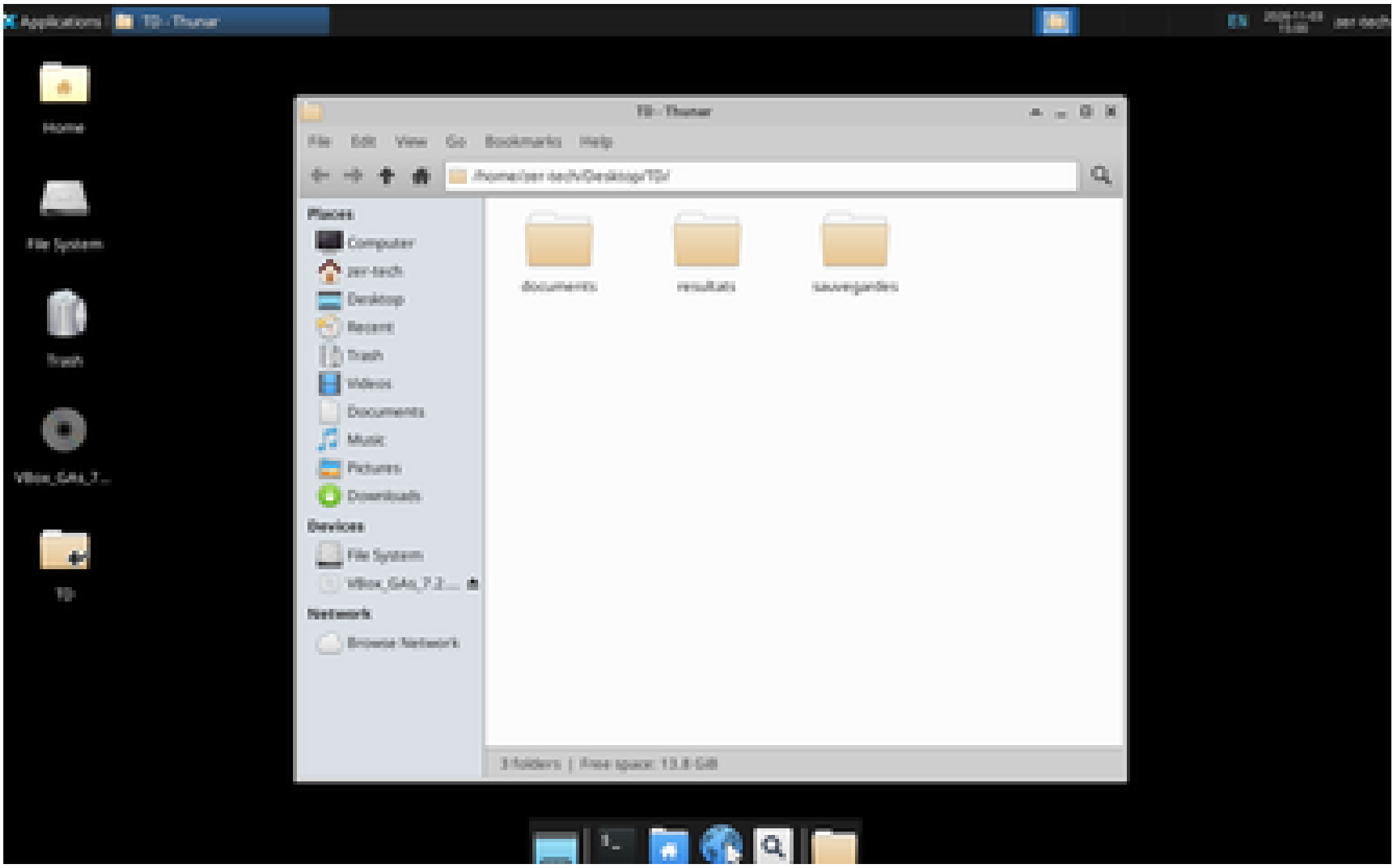

td_linux — Dolphin
home
td_linux
Split
documents
resultats
sauvegardes
3 folders
7:51 PM
11/2/26
7:54 PM
11/2/26



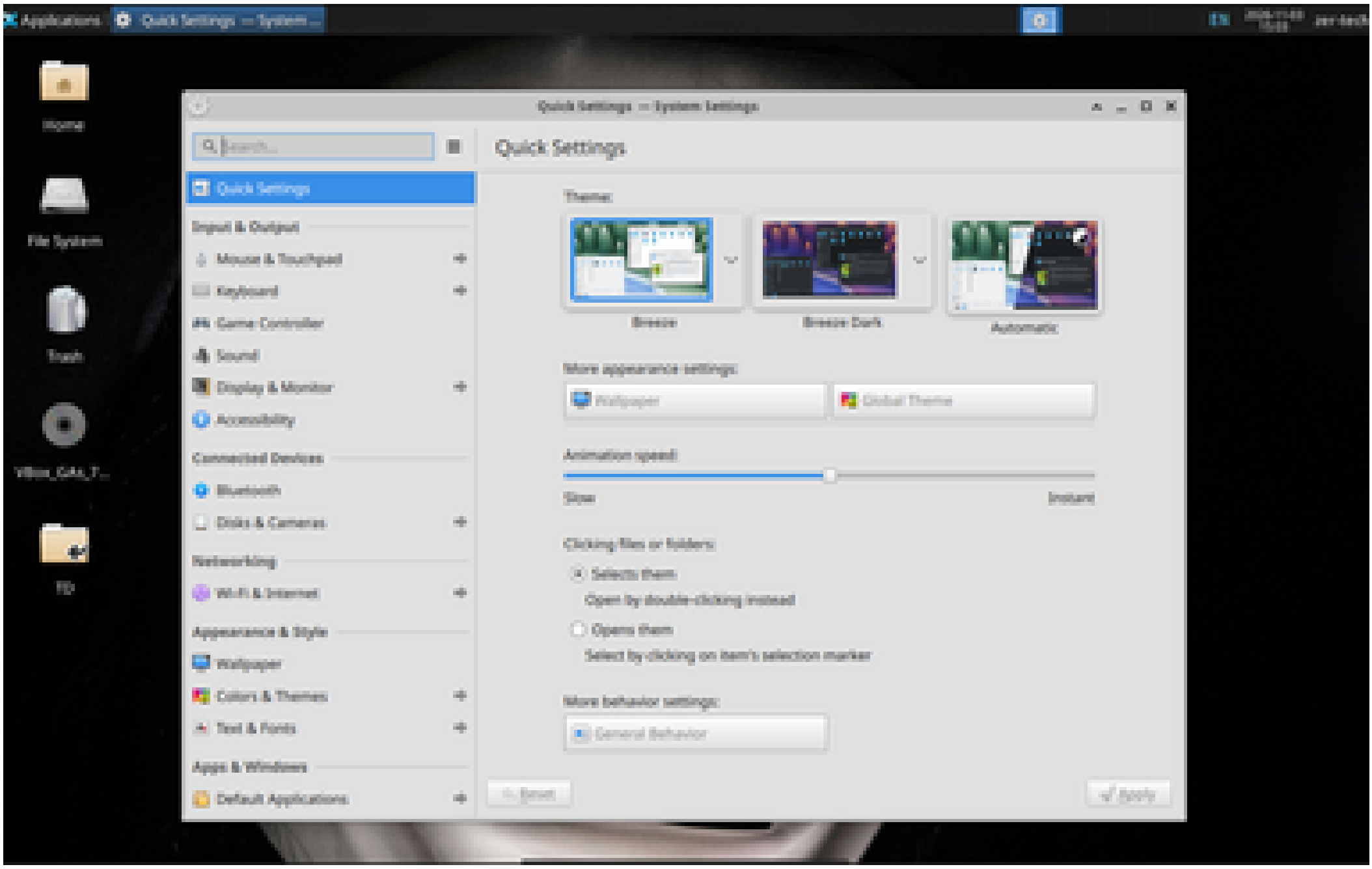

Quick Settings — System Settings
Search...
Quick Settings
Quick Settings
Input & Output
Mouse & Touchpad
Keyboard
Game Controller
Sound
Display & Monitor
Accessibility
Connected Devices
Bluetooth
Disks & Cameras
Networking
Wi-Fi & Internet
Appearance & Style
Wallpaper
Colors & Themes
Text & Fonts
Quick Settings
Theme:
Breeze
Breeze Dark
Automatic
More appearance settings:
Wallpaper
Global Theme
Animation speed:
Slow
Instant
Clicking files or folders:
Selects them
Open by double-clicking instead
Opens them
Select by clicking on item's selection marker
More behavior settings:
General Behavior
Reset
Apply


zer-tech
Password
Ubuntu
GNOME
GNOME Classic
Default Xsession
Plasma (Wayland)
Ubuntu
Xfce Session
Xfce Session (Wayland)


Applications
zer-tech - Thunar
Home
File System
zer-tech - Thunar
File Edit View Go Bookmarks Help
Places
Computer
zer-tech
Desktop
Recent
Trash
Videos
Documents
Music
Pictures
Downloads
Devices
File System
Network
Desktop
Documents
Downloads
Music
Pictures
Public
snap
tst_linux
Templates
Videos
19 folders | 7 files: 12.6 KiB (12,910 bytes) | Free space: 13.8 GiB


Applications
TD - Thunar
Home
File System
Trash
TD
TD - Thunar
File Edit View Go Bookmarks Help
/home/zer-tech/Desktop/TD/
Places
Computer
zer-tech
Desktop
Recent
Trash
Videos
Documents
Music
Pictures
Downloads
Devices
File System
Network
Browse Network
documents
resultats
sauvegardes
3 folders | Free space: 13.8 GB


Applications
Home
File System
Trash
TD



Critère	KDE Plasma	XFCE
Terminal Utilisé	konsole	xfce4-terminal
Gestionnaire de fichiers	Dolphin	Thunar
Accès aux paramètres	systemesetting	xfce-4-settings-manager
personnalisation	Tout est modifiable depuis l’interface sans avoir besoin de faire recours à un outil tiers	personnalisation limitée, simple rapide consomme peu de ressources
mémoire disponible	1.7gi	2.2gi
choix pour un poste donné	poste de travail avec des matériels récents, avec confort visuel au max	poste de travail pour des matériels bas de gamme ou ancien, pour limiter la consommation de RAM

Conclusion

En conclusion, nous appris à manipuler le terminal sur linux et effectuer certaine tâches comme créer et manipuler des dossiers, créer des archives sans avoir besoin d'utiliser l'interface graphique. Ce qui fait la force et la maniabilité extrême de linux.

L'autre point important de ce TD c'est aussi l'initialisation aux packets, nous avons appris a simuler leur installation et suppression via *apt -s install*. la commande *sudo apt install*, collectionne les dépendances des packets et les installe d'un coup.

Difficile de passer a cote de l'étape visant à configurer les deux espaces de travail KDE plasma et XFCE, pendant l'installation nous avons dû changer l'affichage par défaut au dépend de GDN3 pour pouvoir utilisé les différentes affichages.

Les difficultés que nous avons rencontré lors de ce travail était principalement dans l'exercice 2 dans laquelle nous avons créé le fichier services.txt via la commande: *cat > documents/services.txt <<'EOF' web;actif ssh;actif impression;inactif sauvegarde;actif web;actif EOF*. mais malheureusement nous avons dû opter pour une autre option vue celle-ci était défaillante.
Nous avons aussi rencontré quelques difficultés mineurs par rapport a l'affichage de KDE plasma. et les packets cmatrix, neofetch,ncal et cawsay était absent nous les avons installé .

En somme le travail est réussi, et nous sommes en mesure de travailler sur différents espace de travail soit Ubuntu, DKE plasma ou XFCE.